Notes on Gravity

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Preface

General Relativity is one of the most successful theories of physics in history. It seems to hold up to every experimental test thrown at it, even more than a century after its conception. In undergraduate lectures this topic is often presented alongside an introduction to tensor calculus and differential geometry. In contrast, these notes will not contain this background information. The aim is to focus on the consequences of the theory rather than provide a pedagogical overview, and will therefore assume knowledge of some Riemannian geometry. One reference for these notes is the incredibly comprehensive book by Straumann [Str13], though only a small fraction of the content will be covered here. Chapter 1 will recap the theory of General Relativity in a hopefully succinct manner. Chapters 2 & 3 will be devoted to spherically symmetric and rotating black hole solutions. For 4 the aim is to understand the nature of Lorentzian geometry in contrast to the Riemannian case. Witten has some very approachable lectures [Wit20] as does Galloway [Gal14]. The focus will be on the causal structure of spacetime and singularity theorems. These notes are a work in progress, so there will be comments spotted throughout for areas which need to be filled in. Time permitting, I aim to keep this as a long term project so that I can discuss more exotic features of gravity as my understanding improves.

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Chapter 1

Setup of General Relativity

1.1 Motivation

This section will provide an overview of the framework of Einstein's theory. The starting point is to understand the theory of special relativity. We assume that the three dimensions of space and one dimension of time can be treated as a combined space, $\mathbb{R}^4$, equipped with an inner product or metric η. The key distinction between ordinary Euclidean inner product spaces and those discussed here is that the metric is no longer positive definite. The metric can be put in diagonal form and the number of $\pm$ eigenvalues determine its signature. We say that a Lorentzian metric has signature (p, q) if there are p positive eigenvalues and q negative eigenvalues. In the context of special relativity we demand that the metric has signature $(1, 3)$ or $(3, 1)$. Both of the above choices are equivalent, but we will choose to work with the second, so the metric has diagonal form $(-1, +1, +1, +1)$. We write $\mathbb{R}^{1,3}$ or $\mathbb{R}^{3,1}$ when referring to four dimensional Cartesian space equipped with a metric of signature $(3, 1)$ or $(1, 3)$. Roughly speaking, the indefinite signature allows us to distinguish a 'time direction' in spacetime based on the signature. For example, a vector $v \in \mathbb{R}^{1,3}$ is 'pointing in the time direction' if $\eta(v, v) < 0$ in our choice of signature. In fact, the vector could be pointing 'up' in the time axis or 'down'. We need to prescribe which direction time 'flows' in, and this is given by a time-orientation. In this case it is just choice a constant vector, T, with $\eta(T, T) < 0$.

Definition 1.1.1. $\mathbb{R}^{1,3}$ *equipped with the* ***Minkowski metric****,* $\eta(v, v) = -c^2 v_0^2 + \sum_{i=1}^{3} v_i^2$*, and a time-orientation is* ***Minkowski spacetime****.*

With any metric we can associate the orthogonal group $\mathrm{O}(p, q)$, in this case $\mathrm{O}(3, 1)$, which is the group linear transformations on $\mathbb{R}^{1,3}$ which preserve η. This is the **Lorentz group**. Not all of these transformations are considered to be physical. We restrict to the **proper orthochronous** Lorentz group $\mathrm{SO}^+(3, 1)$ which only includes those transformations which have positive determinant and do not allow time and space directions to be reflected. This is also denoted $\mathrm{SO}_0(3, 1)$ for the connected component of the identity. In special relativity observers moving at constant velocity define a so-called inertial reference frame. Any other inertial frame is related to by a transformation in the above group. The transformations can also act vectors with $\eta(v, v) = 0$. In two dimensions, $\mathbb{R}^{1,1}$, with time on the y-axis and space on the x, this relation defines two lines crossing the origin with gradient $\pm 1/c$; in other words light rays with velocity c. In this way we recover one of the postulates of special relativity that the speed of light must be constant in all frames of reference.

To get to General Relativity (GR) we suppose that spacetime is no longer described by a Cartesian space, but has the property of being curved. To enforce the equivalence principle we demand that no matter how curved spacetime is, it must locally look flat. This leads to considering that spacetime must be described by a manifold, M, instead of a vector space. It also ought to be a smooth manifold to permit the methods of calculus to be used. Moreover, we want that special relativity should hold locally at each point in space, so it should 'look like' Minkowski spacetime at any point. From Riemannian geometry we know that a manifold at a point looks like its tangent space at that point, the content of the normal neighbourhood theorem. It makes sense to then demand that our tangent spaces are actually Minkowski spaces. This idea is part of Cartan geometry.

In the physics literature, tensors are often **defined** by their transformation law. By contrast, in mathematics, one introduces tensors as arbitrary sections of the appropriate tensor tangent bundle. These two viewpoints don't agree a priori. For example, even in Minkowski space, a physicist only considers special 'four-vectors' to be vectors. To a mathematician, any old tuple of four numbers in $\mathbb{R}^4$ is a perfectly well-defined vector. However, in physics, not even all real numbers are classed as 'scalar' quantities. For example, the energy of a particle changes depending on which reference frame one chooses, and hence is not considered a scalar. These viewpoints can be reconciled by being slightly more pedantic. A postulate of special relativity is that that physical laws should not depend on the choice of inertial reference frame. This can be enforced by demanding that physical quantities should be described by representations of the appropriate group. For example, vectors in Minkowski space which transform in the Lorentz group are **four-vectors**.

In GR, we need to promote this to the manifold level. We want a metric tensor defined pointwise as, $g_p : T_pM \times T_pM \to \mathbb{R}$, which has a $(3,1)$-signature. Recall that there always exists a **frame bundle** on any manifold which consists of the disjoint union of the set of bases at each point. More precisely, this is a principal $\mathrm{GL}(4,\mathbb{R})$-bundle. If there exists a Lorentzian metric, there is a left action of Lorentz group on the frame bundle defined by a representation. We can then construct the assiocated bundle to a given representation with fibre isomorphic to $\mathbb{R}^4$. The upshot is then that tensor fields ought to sections of the associated bundle, to ensure that they transform correctly. The usual tangent bundle to a manifold can be constructed from the default structure group $\mathrm{GL}(4,\mathbb{R})$. Here we have a reduction of the structure group to the proper orthochronous Lorentz group, encoding the necessary transformation properties on the associated bundle.

Now, with a Lorentzain metric, vectors can be split into three classes.

Definition 1.1.2. *At a point $p \in M$ a vector $X \in T_pM$ is said to be:*

1. ***Timelike*** *if $g(X,X) < 0$.*
2. ***Lightlike*** *if $g(X,X) = 0$.*
3. ***Spacelike*** *if $g(X,X) > 0$.*

A Lorentzian metric on a spacetime also defines light cones, $\mathcal{V}_p$, in the tangent spaces at points by,

$$\mathcal{V}_p \coloneqq \{X \in T_pM : g_p(X,X) = 0\}.$$

At each point there are actually two 'components' of $\mathcal{V}_p$, and we need to distinguish one of these as future pointing. This assignment needs to smooth and defined over the whole spacetime to have a reasonable sense of the 'future'. Such an assignment is a time-orientation.

Definition 1.1.3. *A Lorentzian manifold M is time-orientable if it admits a smooth, timelike, nowhere vanishing global section $T \in \mathfrak{X}(M)$.*

Using the time-orientation we can define whether things are travelling to the future or the past. A vector $X \in T_pM$ is **future directed** at a point $p \in M$ if $g_p(X,T) < 0$. Note that any other timelike, global, nowhere vanishing section is still a valid time-orientation. If we're being proper then we should quotient out by the equivalence relation

$$T, T' \in \mathfrak{X}(M), \quad T \sim T' \iff g(T,T') < 0,$$

so that then a choice of time-orientation is techinically an equivalence class $[T]_\sim$. Finally, orientability is imposed as a condition for all spacetimes we examine in these notes. Collecting all of this information gives a working definition.

Definition 1.1.4. ***Spacetime*** *is tuple $(M, g, \nabla, \mathrm{d}\mu_g, T)$ where:*

1. *M is a (smooth) connected four-dimensional manifold;*
2. *$g \in \mathrm{Sym}^2(T^\star M)$ is a Lorentzian metric with signature $(-,+,+,+)$;*
3. *∇ is the Levi-Civita connection associated to g;*
4. *$\mathrm{d}\mu_g \in \Omega^4(M)$ is a volume form on M;*
5. *and $T \in \mathfrak{X}(M)$ is a time-orientation.*

Recall that every Riemannian manifold admits a unique connection which is metric compatible and torsion free. The same is true for a Lorentzian manifold, and this is still called the Levi-Civita connection. In more general theories of gravity one considers arbitrary connections which are perhaps not torsion free etc. This is why we've made the Levi-Civita connection explicit in our definition despite it coming for free with a choice of metric. On the other hand not all manifolds are orientable, this is an extra condition we impose on a valid spacetime. Furthermore, we have the following statements.

Remark 1.1.1. *A smooth manifold may not admit a Lorentzian metric, despite always admitting a Riemannian one.*

Remark 1.1.2. *An orientable Lorentzian manifold may not admit a time-orientation.*

The spacetimes which are allowed is restricted somewhat by the above two facts. More about causality and Lorentzian geometry is discussed in Chapter 4.

1.2 Einstein-Hilbert Action

General relativity is a classical field theory where the dynamical field is the metric tensor. This is a crucial difference from other forces modelled by quantum gauge theories, which take place on a fixed background. Einstein originally deduced the equations of motion from physical principles. Another route which was taken by Hilbert —[and maybe also Einstein, check]— is to guess an appropriate Lagrangian and derive the equations of motion from the principle of least action.

Definition 1.2.1. *The* ***Einstein-Hilbert action*** *on a Spacetime M is:*

$$S[g] = \int_M \star \left(\frac{1}{2\kappa} R + \mathcal{L}_{\mathcal{M}} \right) \tag{1.1}$$

1. $\kappa = 8\pi G c^{-4}$. *Where G is the gravitational constant and c is the speed of light;*
2. $\star : \Omega^k(M) \to \Omega^{n-k}(M)$ *is the Hodge dual for the metric g;*
3. *R is the Ricci scalar;*
4. $\mathcal{L}_{\mathcal{M}}$ *is a matter Lagrangian describing the fields and particle distribution.*

It's natural to ask why this specific Lagrangian and not another. One could just say this mathematical formula is the one that gets the correct equations of motion, but there are multiple ways to justify it from physical principles. One such justification is that this action is actually the unique low energy effective gravitational field theory [**?**]. From here on we will make use of local coordinates on spacetime when needed. Denote the chart around a neighbourhood $p \in U \subseteq M$ by $x^\mu \in \mathbb{R}^4$ where $\mu = \{0, 1, 2, 3\}$. Correspondingly, the tangent space basis vectors will be denoted $\{\partial_\mu\}$ and the cotangent basis $\{\mathrm{d}x^\mu\}$. The Einstein summation convention will always be assumed unless stated otherwise. The gravitational constant and the speed of light will be also be set to unity in the appropriate units.

Definition 1.2.2. *The energy momentum tensor, $T_{\mu\nu}$, is defined in terms of the matter Lagrangian by*

$$T_{\mu\nu} := \frac{-2}{\sqrt{-g}} \frac{\delta}{\delta g^{\mu\nu}} \left[\sqrt{-g} \mathcal{L}_{\mathcal{M}} \right]. \tag{1.2}$$

The energy momentum tensor encapsulates the fact matter and energy act as a source for gravity. This quantity won't be used much in these notes, but it is obviously of extreme importance since there is matter in the universe! Moreover, the energy momentum tensor also includes contributions from 'fields' such as the electromagnetic field or scalar fields as well as matter. In a certain sense everything gravitates; even gravity. This is one of the reasons that general relativity is so difficult.

Theorem 1.2.1. *Let M be a spacetime with no boundary and S the Einstein-Hilbert action. The sationary action principle, $\delta S = 0$, implies Einstein's field equations in local coordinates (x, U):*

$$R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} = \kappa T_{\mu\nu} \tag{1.3}$$

Proof. The variation of the matter Lagrangian is almost by definition the energy momentum tensor so we'll focus on only the first term of the Lagrangian density.

$$\delta\left(\sqrt{-g}R\right) = \delta\left(\sqrt{-g}g^{\mu\nu}R_{\mu\nu}\right) = \delta\left(R_{\mu\nu}\right)\sqrt{-g}g^{\mu\nu} + \delta\left(\sqrt{-g}g^{\mu\nu}\right)R_{\mu\nu} \tag{1.4}$$

Let's compute the second term first. Consider the variation of the inverse metric $\delta(g^{\mu\nu})$. We start by using the definition of the inverse,

$$g^{\mu\sigma} g_{\sigma\nu} = \delta^\mu_\nu, \tag{1.5}$$

now apply the variation to both sides and rearrange:

$$\begin{aligned}
&\delta(g^{\mu\nu}g_{\nu\sigma}) = \delta(g^{\mu\nu})g_{\nu\sigma} + g^{\mu\nu}\delta(g_{\nu\sigma}) = 0\\
&\delta(g^{\mu\nu})g_{\nu\sigma} = -g^{\mu\nu}\delta(g_{\nu\sigma})\\
&\delta(g^{\mu\nu})g_{\nu\sigma}g^{\sigma\rho} = -g^{\mu\nu}\delta(g_{\nu\sigma})g^{\sigma\rho}\\
&\delta(g^{\mu\rho}) = -g^{\mu\nu}\delta(g_{\nu\sigma})g^{\sigma\rho}
\end{aligned}$$

Next observe we need to work out the variation of the determinant of a metric. This follows from Jacobi's formula,

$$\delta(g) = g\mathrm{Tr}(g^{\mu\alpha}\delta(g_{\alpha\nu})), \tag{1.6}$$

where we can use the chain rule to obtain,

$$\begin{aligned}
\delta\sqrt{-g} &= \frac{1}{2\sqrt{-g}}(-g)g^{\mu\nu}\delta(g_{\mu\nu})\\
\delta\sqrt{-g} &= \frac{1}{2}\sqrt{-g}g^{\mu\nu}\delta(g_{\mu\nu}).
\end{aligned}$$

We can simplify the calculation of $\delta R_{\mu\nu}$ by using Riemann normal coordinates at $p \in U$, meaning the Christoffel symbols vanish, leaving only

$$\delta R_{\mu\nu} = \partial_\alpha\left(\delta\Gamma^\alpha_{\mu\nu}\right) - \partial_\nu\left(\delta\Gamma^\alpha_{\mu\alpha}\right). \tag{1.7}$$

We can now swap $\partial \to \nabla$ since $\delta\Gamma$ is actually a tensor quantity. This result is the Platini identity.

$$\delta R_{\mu\nu} = \nabla_\alpha\delta\Gamma^\alpha_{\mu\nu} - \nabla_\nu\delta\Gamma^\alpha_{\mu\alpha} \tag{1.8}$$

Then we can also write

$$g^{\mu\nu}\delta R_{\mu\nu} = \nabla_\alpha g^{\mu\nu}\delta\Gamma^\alpha_{\mu\nu} - \nabla_\nu g^{\mu\nu}\Gamma^\alpha_{\alpha\mu} = \nabla_\alpha W^\alpha,$$

where the vector field is $W^\alpha = g^{\mu\nu}\delta\Gamma^\alpha_{\mu\nu} - g^{\alpha\nu}\delta\Gamma^\mu_{\mu\nu}$. Now we can return to the full expansion:

$$\begin{aligned}
\delta S = \frac{1}{2\kappa}\int_M \delta(\star R) &= \frac{1}{2\kappa}\int_M \star\left(\frac{1}{2}g^{\mu\nu}R\delta g_{\mu\nu} - g^{\mu\alpha}g^{\beta\nu}R_{\mu\nu}\delta g_{\alpha\beta}\right) + \mathrm{d}\star W^\flat\\
&= \frac{1}{2\kappa}\int_M \star\delta g_{\mu\nu}\left(\frac{1}{2}Rg^{\mu\nu} - R^{\mu\nu}\right) + \int_{\partial M}\star W^\flat
\end{aligned}$$

Assuming that $\partial M = \varnothing$ or specifying appropriate boundary conditions gives the desired result. □

—[MORE ON BDRY]—. The field equations are incredibly difficult to solve. They are non-linear, 2nd order, coupled partial differential equations in terms of the metric. The only way to get analytic solutions is in highly symmetric toy models, as we will see in the next two chapters. Yet even then it can be cumbersome! One way to simplify the situation is to consider the case of a vanishing energy momentum tensor

$$\begin{aligned}
R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} &= 0\\
\implies R - \frac{1}{2}R\delta^\mu_\mu &= 0\\
\implies R &= 0,
\end{aligned}$$

then substitute $R = 0$ back into the vacuum field equations to get,

$$R_{\mu\nu} = 0, \tag{1.9}$$

which is the common form Einstein's vacuum field equations. These notes will be focused mostly on this restricted regime.

1.3 Geodesic Flow

We can model the motion of small objects travelling in spacetime in the absence of forces with geodesics. Since the mass is assumed small, we can ignore the back reaction of the mass of the matter on the spacetime/gravitational field. Recall that a **symplectic form** is a closed non-degenerate 2-form. A manifold with equipped with a symplectic form is a **symplectic manifold**. When such a space also has a designated a smooth energy function, it is a Hamiltonian system. It will be useful to formalise the dynamics here as a Hamiltonian system on the phase space of spacetime, $T^\star M$. We may also make use of the bundle isomorphism $T^\star M \cong TM$ induced by the metric as $(x^\mu, p_\mu) \mapsto (x^\mu, g^{\mu\nu} p_\nu)$. This lets us pull back the symplectic form to the tangent bundle as,

$$\omega_g = \mathrm{d}x^\mu \wedge \mathrm{d}\left(g_{\mu\nu} p^\nu\right),$$

and then one can consider Lagrangian mechanics. We will stick to Hamilton's formulation for these upcoming definitions.

Definition 1.3.1 (Geodesic Flow)**.** *For any Riemmanian manifold (M, g), we have a Hamiltonian system, $(T^\star M, \omega, H)$, where in local coordinates (x^μ, p_μ) we define:*

1. *The **symplectic form**, $\omega \in \Omega^2(T^\star M)$, in local coordinates is given by $\omega = \mathrm{d}x^\mu \wedge \mathrm{d}p_\mu$.*
2. *The Hamiltonian is $H = \frac{1}{2} g^{\mu\nu} p_\mu p_\nu$.*

Definition 1.3.2. *Consider a Hamiltonian system $(T^\star M, \omega, H)$ on spacetime. $\exists!\ X_H \in \mathfrak{X}(T^\star M)$ called the **Hamiltonian vector field** satisfying*

$$\iota_{X_H} \omega = \mathrm{d}H, \tag{1.10}$$

and defining Hamilton's equations of motion.

Now, for geodesic flow, the symplectic form is just the canonical one. We need a conserved energy to define a Hamiltonian system, here it is just the usual energy defined from the square of momentum, except using the induced metric. We will write trajectories in spacetime as $x^\mu(\tau)$ as a slight abuse of notation for the coordinates of the curve in a chart, $(x^\mu \circ \gamma)(\tau)$. The components of the tangent vector to the curve at a point is then written $\dot{x}^\mu \partial_\mu$.

Theorem 1.3.1. *Let (M, g) be a Riemannian manifold. Geodesic flow is satisfied by an appropriately paramaterised curve $\gamma : \mathbb{R} \longrightarrow M$ if*

$$\nabla_{\dot\gamma} \dot\gamma = 0 \tag{1.11}$$

$$\iff \ddot{x}^\mu + \Gamma^\mu_{\sigma\nu} \dot{x}^\sigma \dot{x}^\nu = 0, \tag{1.12}$$

for all charts (x, U) where $x^\mu(\tau)$ denotes the local coordinates of the curve.

Proof. Start with Hamilton's equations of motion in Definition 1.3.2, which is equivalent to

$$\dot{x}^\mu = \frac{\partial H}{\partial p_\mu}, \quad \dot{p}_\mu = -\frac{\partial H}{\partial x^\mu}.$$

Then we obtain,

$$\dot{x}^\mu = g^{\mu\nu} p_\nu, \quad \dot{p}_\mu = -\frac{1}{2}\partial_\sigma(g^{\mu\nu}) p_\mu p_\nu.$$

Now take the derivative of the left-hand equation to get,

$$\begin{aligned} \ddot{x}^\mu &= g^{\mu\nu}\dot{p}_\nu + \dot{x}^\sigma \partial_\sigma g^{\mu\nu} p_\nu \\ &= g^{\mu\nu}\dot{p}_\nu + \dot{x}^\sigma \partial_\sigma g^{\mu\nu} g_{\nu\rho}\dot{x}^\rho, \end{aligned}$$

where we've used the chain rule $\frac{\mathrm{d}}{\mathrm{d}\tau} = \dot{x}^\sigma \partial_\sigma$, and the definition of four-momentum $p_\nu = g_{\mu\rho}\dot{x}^\rho$. Next substitute the equation for $\dot{p}_\nu$ to get,

$$\ddot{x}^\mu = -\frac{1}{2}\partial_\nu g^{\gamma\delta} p_\gamma p_\delta g^{\mu\nu} + \dot{x}^\sigma \partial_\sigma g_{\nu\rho}\dot{x}^\rho.$$

We need to eliminate the derivative of the inverse metric so note from product rule we have :

$$\begin{aligned} \partial_\gamma(g_{\mu\nu}g^{\nu\sigma}) &= \partial_\gamma(\delta^\sigma_\mu) = 0 \\ &= g_{\mu\nu}\partial_\gamma g^{\nu\sigma} + g^{\nu\sigma}\partial_\gamma g_{\mu\nu} = 0 \\ \implies g^{\mu\rho} g_{\mu\nu}\partial_\gamma g^{\nu\sigma} &= -g^{\mu\rho} g^{\nu\sigma}\partial_\gamma g_{\mu\nu} \\ \implies \partial_\nu g^{\gamma\delta} &= -g^{\mu\gamma} g^{\sigma\delta}\partial_\nu g_{\mu\sigma}. \end{aligned}$$

Using this identity we obtain,

$$\ddot{x}^\mu = \frac{1}{2} g^{\mu\gamma} g^{\sigma\delta}\partial_\nu g_{\mu\sigma} g_{\beta\gamma}\dot{x}^\beta g_{\alpha\delta}\dot{x}^\alpha g^{\mu\nu} - g^{\alpha\mu} g^{\beta\nu}\partial_\sigma g_{\alpha\beta} g_{\nu\rho}\dot{x}^\sigma \dot{x}^\rho$$

$$\implies \ddot{x}^\mu = \frac{1}{2} g^{\mu\nu}\dot{x}^\sigma \dot{x}^\alpha \partial_\nu g_{\alpha\sigma} - g^{\alpha\mu}\dot{x}^\sigma \dot{x}^\beta \partial_\sigma g_{\alpha\beta}$$

$$\ddot{x}^\mu = \frac{1}{2} g^{\mu\alpha}(\partial_\alpha g_{\sigma\nu} - \partial_\sigma g_{\alpha\nu} - \partial_\nu g_{\sigma\alpha})\dot{x}^\sigma \dot{x}^\nu$$

$$\implies \ddot{x}^\mu + \Gamma^\mu_{\sigma\nu}\dot{x}^\sigma \dot{x}^\nu = 0,$$

where we've relabelled indices and recognised the formula for the Christoffel symbols of a Levi-Civita connection. On the other hand, by the definition of the induced covariant derivative on a curve

$$\begin{aligned} \nabla_{\dot{\gamma}}\dot{\gamma} = 0 &\iff \dot{x}^\mu \nabla_{\partial_\mu}\dot{x}^\nu = 0 \\ \iff \frac{\mathrm{d}x^\mu}{\mathrm{d}\tau}\frac{\partial}{\partial x^\mu}\frac{\mathrm{d}x^\nu}{\mathrm{d}\tau} &+ \frac{\mathrm{d}x^\mu}{\mathrm{d}\tau}\Gamma^\mu_{\sigma\nu}\frac{\mathrm{d}x^\sigma}{\mathrm{d}\tau} = 0. \end{aligned}$$

Finally, by the chain rule, we obtain

$$\ddot{x}^\mu + \Gamma^\mu_{\sigma\nu}\dot{x}^\sigma \dot{x}^\nu = 0.$$

□

Note that we're being somewhat informal in distinguishing between geodesic and so-called co-geodesic flow, since the equations of motion are the same it won't make a difference in these notes. We will identify $g^{\mu\nu}p_\nu = p^\mu := \dot{x}^\mu$ as the **four-momentum**. For any path of a massive particle in spacetime, geodesic or not, we can define the length by integration of the metric.

Definition 1.3.3. *The **length** of curve $\gamma : [a, b] \to M$ in spacetime is defined as:*

$$L(\gamma) := \int_a^b \sqrt{|g(\dot{\gamma}(\lambda), \dot{\gamma}(\lambda))|}\mathrm{d}\lambda$$

In the case of a massive particle we can use this to define the **proper time** elapsed along a trajectory.

Definition 1.3.4. *The **proper time** τ for a massive particle is the arc length parameter associated to the length functional.*

Traditionally this is written as $ds^2 = -d\tau^2$. Recall that in Riemannian geometry geodesics are stationary points of the length functional which locally minimise distances (but not necessarily globally). In Lorentzian geometry geodesics are still stationary points of the new length functional, but now we have the interpretation of locally **maximising** elapsed proper time. Intuitively in the case of special relativity, a stationary particle relative to a frame has no time dilation or anything strange. On the other hand wiggling around causes less proper time to elapsed due to time dilation effects. A similar story is true for general relativity, infinitesimally at least. Note that the concept of things being at rest is a lot more subtle in general relativity, since inertial frames can only be defined locally.

Definition 1.3.5. *A **conserved quantity**, $f \in C^\infty(T^\star M)$, is constant along geodesic flow,*

$$X_H(f) = \gamma_\star \partial_\tau \cdot f = \frac{\mathrm{d}}{\mathrm{d}\tau}(f \circ \gamma)(\tau) = 0, \tag{1.13}$$

where $\gamma(\tau) = (x^\mu(\tau), p_\mu(\tau))$ is now a curve in phase space, with the usual abuse of notation for coordinates on the image.

Definition 1.3.6. *A vector field $X \in \mathfrak{X}(T^\star M)$ is a (Hamiltonian) **symmetry** for a system $(T^\star M, \omega, H)$ if:*

1. *$X(H) = 0$;*
2. *$\mathcal{L}_X \omega = 0$.*

Theorem 1.3.2 (Noether)**.** *If $X \in \mathfrak{X}(T^\star M)$ is a Hamiltonian symmetry and $H^1_{\mathrm{dR}}(T^\star M; \mathbb{R}) = 0$ then $\exists f \in C^\infty(T^\star M)$ such that*

$$\iota_X \omega = \mathrm{d}f, \tag{1.14}$$

where f is a conserved quantity.

Proof. To show such a function exists we just expand the Lie derivative using Cartan's magic formula,

$$\mathcal{L}_X \omega = \iota_X \circ \mathrm{d}\omega + \mathrm{d} \circ \iota_X \omega = 0,$$

but we know that $\mathrm{d}\omega = 0$ since ω is symplectic. Then remaining is,

$$\mathrm{d} \circ \iota_X \omega = 0,$$

which implies, using the triviality of $H^1(T^\star M)$, that $\exists f \in C^\infty(M)$ satisfying $\iota_X \omega = \mathrm{d}f$. Now it remains to show that f is conserved along flows generated by X. We just have to apply the interior product successively to the symplectic form,

$$\begin{aligned} \iota_X \circ \iota_{X_H} \omega = \iota_X \circ \mathrm{d}H = X(H) = 0 \\ -\iota_{X_H} \circ \iota_X \omega = -\iota_{X_H} \circ \mathrm{d}f = 0 \implies X_H(f) = 0. \end{aligned}$$

□

The above is Noether's theorem in Hamiltonian mechanics, which we will make use of this when solving the geodesic equations. The existence of conserved quantities lets us simplify the equations of motion. If there exists a set of n 'independent' such quantities then we say the system is completely solvable. More specifically we can introduce the following definitions.

Definition 1.3.7. *For a Hamiltonian system $(T^\star M, \omega, H)$ define the* ***Poisson bracket*** *as the bilinear map:*

$$\{-,-\} : C^\infty(T^\star M) \ni (f,g) \longmapsto \{f,g\} \coloneqq \omega(X_f, X_g) \in C^\infty(T^\star M) \tag{1.15}$$

The Poisson bracket is equivalentley given by $\{f,g\} = X_g f$, and hence the adjoint action is a derivation on the algebra of smooth functions. Moreover, the bracket is alternating, and the non-degeneracy of the symplectic forms makes it satisfy the Jacobi identity, giving a Lie algebra structure. The time evolution of an observable is given by $\{-, H\}$, and so conserved quantities necessarily commute with the Hamiltonian. This brings us to a concrete definition of what it means for geodesic flow to be solvable.

Definition 1.3.8. *A Hamiltonian system $(T^\star M, \omega, H)$ is* ***Liouville integrable*** *if the following conditions are satisified.*

1. *There exists $n = \dim M$ conserved quantities, $\{f_i\}_{i=1}^n$, which are in* ***involution****:*

$$\{f_i, f_j\} = 0 \ \forall i, j$$

2. *The conserved quantities are functionally independent,*

$$\mathrm{d}f_1 \wedge \ldots \wedge \mathrm{d}f_n \neq 0,$$

on a dense set.

Another bracket operation which exists on any smooth manifold is the Lie bracket. It turns out that the Poisson bracket is related to this existing bracket.

Proposition 1.3.1. *Let $(T^\star M, \omega, H)$ be a Hamiltonian system. Then $\forall f, g \in C^\infty(T^\star M)$ we have:*

$$X_{\{f,g\}} = -[X_f, X_g] \tag{1.16}$$

Proof. Recall that for any function $h \in C^\infty(T^\star M)$ we have an associated vector field $X_h \in \mathfrak{X}(T^\star M)$. Then from the LHS,

$$\begin{aligned}
\iota_{X_{\{f,g\}}}\omega &= \mathrm{d}\{f,g\} = \mathrm{d}(X_g \cdot f) \\
&= \mathrm{d} \circ \mathcal{L}_{X_g} f = \mathcal{L}_{X_g} \circ \mathrm{d} f \\
&= \mathcal{L}_{X_g} \circ \iota_{X_f}\omega \\
&= \iota_{\mathcal{L}_{X_g} X_f}\omega + \iota_{X_f} \circ \mathcal{L}_{X_g}\omega \\
&= \iota_{[X_g, X_f]}\omega,
\end{aligned}$$

and the result follows from the non-degeneracy of the symplectic form. □

Recall that in Riemannian geometry one has the notion of Killing vector fields:

Definition 1.3.9. *A **Killing vector field**, $X \in \mathfrak{X}(M)$, satisfies $\mathcal{L}_X g = 0$.*

Equivalently, a Killing vector field can be thought of as generating infinitesimal isometries along its flow. By studying symmetries of a spacetime one can immediately deduce associated conserved quantities.

Proposition 1.3.2. *Given a geodesic flow on a spacetime M we have that a Killing vector field, $K \in \mathfrak{X}(M)$, induces a conserved quantity, $g(K, X)$, where X is the tangent vector field to the geodesic.*

Proof. First let us derive one of the defining properties of Killing vector fields. By definition, if K is a Killing field then $\forall Y, Z \in \mathfrak{X}(M)$ we have

$$0 = (\mathcal{L}_K g)(Y, Z) = K(g(Y, Z)) - g([K, Y], Z) - g(Y, [K, Z]).$$

Using metric compatibility and the torsion free property of the connection we then expand

$$0 = g(\nabla_K Y, Z) + g(Y, \nabla_K Z) - g(\nabla_K Y - \nabla_Y K, Z) - g(Y, \nabla_K Z - \nabla_Z K),$$

and finally cancelling terms:

$$g(\nabla_Y K, Z) + g(Y, \nabla_Z K) = 0 \tag{1.17}$$

This is known as the **Killing equation**. A more familiar coordinate expression can be also be derived

$$\begin{aligned}
0 = (\mathcal{L}_K g)(\partial_\mu, \partial_\nu) &= g(\nabla_{\partial_\mu} K, \partial_\nu) + g(\partial_\mu, \nabla_{\partial_\nu} K) \\
&= g((\nabla_\mu K^\sigma)\partial_\sigma, \partial_\nu) + g(\partial_\nu, (\nabla_\nu K^\sigma)\partial_\sigma) \\
&= \nabla_\mu K^\sigma g_{\sigma\nu} + g_{\nu\sigma}\nabla_\nu K^\sigma,
\end{aligned}$$

which gives the equivalent equation:

$$\nabla_\mu K_\nu + \nabla_\nu K_\mu = 0 \tag{1.18}$$

Now let X be a tangent vector field to a geodesic. Looking at the conserved quantity,

$$X(g(K, X)) = (\nabla_X g)(K, X) + g(\nabla_X K, X) + g(K, \nabla_X X),$$

the first term is zero by metric compatibility and the third term vanishes since X is a tangent to a geodesic. Finally note that $g(X,X)$ is constant, so we also have

$$\begin{aligned} 0 = (\mathcal{L}_K g)(X,X) &= \mathcal{L}_K(g(X,X)) - 2g(\mathcal{L}_K X, X) \\ &= 2g([X,K],X) = 2g(\nabla_X K - \nabla_K X, X) \\ &= 2g(\nabla_X K, X) - 2g(\nabla_K X, X) \\ &= 2g(\nabla_X K, X) - K(g(X,X)) \\ &\implies g(\nabla_X K, X) = 0, \end{aligned}$$

showing the vanishing of the second term in $X(g(K,X))$ and giving the desired result. □

The key property of Killing vector fields is that they act on configuration space, which is why it is easy to extract conserved quantites from them. In general not all conserved quantities will arise in this fashion from symmetries of spacetime. Some may have a non-trivial action on the whole of phase space, $T^\star M$, in which case deducing their existence and associated conserved quantities can be difficult. These are often called **hidden symmetries**. An example of this occurs for geodesic flow in Kerr spacetime, discussed in Section 3.3.

The above discussion suffices to describe the path a unit mass particle traverses in spacetime. In general we also want to consider massless particles such as photons. In the massive case the Hamiltonian defines the normalisation of momenta by $H = -\frac{1}{2}$, in the massless case the hamiltonian is zero since light rays are null $g(\dot\gamma, \dot\gamma) = 0$. Furthermore, the length function is identiacally zero so we must find a more suitable parameter to describe the distance travelled. We can actually define the paramater as that which satisfies the geodesic equation.

Definition 1.3.10. *Consider a Riemannian manifold (M,g) and a curve $x^\mu(\lambda)$ in a chart (x,U). An **affine parameter**, λ, is a parameter which satisfies the geodesic equation*

$$\frac{\mathrm{d}^2 x^\mu}{\mathrm{d}\lambda^2} + \Gamma^\mu_{\sigma\nu}\frac{\mathrm{d}x^\sigma}{\mathrm{d}\lambda}\frac{\mathrm{d}x^\nu}{\mathrm{d}\lambda} = 0$$

This parameter is unique up to an affine reparameterisation $\lambda' = a\lambda + b$ where $a, b \in \mathbb{R}$. So even along a light ray, there is still a way to track a 'length' along the geodesic.

1.4 General Covariance

—[This section is under construction]— In the abstract mathematical sense a manifold is a perfectly well-defined concept. The extent to which spacetime itself coincides with this definition is not immediatey obvious. When Einstein was originally formulating his theory he struggled with this concept. In this section we'll briefly talk about some of these issues and resolutions.

—[hole argument]—The resolution to the paradox comes from realising that different field configurations have to be deemed equivalent if they are related to each other by an isometry. In other words we must quotient the space of metrics by some equivalence relation.

Definition 1.4.1. *Two spacetimes (M,g) and (M',g') are **equivalent** if they are isometric. More precisely, there exists a diffeomorphism $\varphi : M \longrightarrow M'$ such that $\varphi^\star g' = g$.*

Definition 1.4.2. *A spacetime M is **extendible** to M' if there is a proper open set $N \subsetneq M'$ such that M is isometric to N.*

—[move this definition to the singularity section in ch4]—.

1.5 Moving Frames

1.5.1 Klein Geometry of Minkowski Space

This section will give introduction to Cartan geometry in the context of GR. An exposition can be found in [Cat15] and [Wis10]. A full overview of the geometry is in [Sha02]. In Minkowski spacetime there is a group of symmetries which preserves the metric, the isometry group. In particular, we have translational invariance and Lorentz invariance. Combining these two together gives us the isometry group of Minkowski space; the **Poincaré group** $\mathrm{ISO}(3,1)$. We denote the connected component of the identity of the Lorentz group by $\mathrm{SO}_0(3,1)$, and likewise for the Poincaré group $\mathrm{ISO}_0(3,1)$. We may get sloppy when referring to the Lorentz group as opposed to the connected component of the identity. Both subgroups of the Poincaré group have natural right actions on the whole group itself. We can study the coset spaces of these actions to understand geometric aspects of Minkowski spacetime.

Consider the action of the three-dimensional Euclidean Poincaré group $\mathrm{ISO}(3)$ on the connected Minkowski Poinacaré group. The coset space $\mathrm{ISO}_0(3,1)/\mathrm{ISO}(3)$ is the stabiliser of this action. Hence, this coset space characterises the set of **spacelike hyperplanes** in Minkowski spacetime. On the other hand consider the action of connected Lorentz group on the connected Poincaré group. In this case we're 'modding out' by all rotations and boosts, hence the stabilisers are just points. The coset space $\mathrm{ISO}_0(3,1)/\mathrm{SO}_0(3,1)$ characterises all the **events** in Minkowski spacetime.

The above exaples are special cases of a general idea by Felix Klein.

Definition 1.5.1. *A **Klein geometry**, (H,G), consists of a Lie group G and a closed subgroup H such that the coset space is connected.*

Sometimes this definition is relaxed by allowing for disconnected Klein geometries, we will not consider these here. Note that from Lie theory we know that a closed subgroup of a Lie group is automatically a Lie subgroup. Furthermore, the coset space $(G/H = X)$ is a smooth manifold, which can be acted on transitatively on the left by G with group multiplication. In this case the pair G, X has X as a **homogenous space**. The subgroup also Hhas the natural right action on G induced from group multiplication. One can use this action to construct a bundle over the coset space $G/H$with fibres as left cosets of H. Now the right action is free and transitative, so the coset space is a **principal homogenous space** or **H-torsor**. In this case the fibre bundle is actually a **principal H-bundle** over G/H.

Recall that Lie group always has an adjoint left action on its Lie algebra. Equivalently we have a representation of H on the underlying vector space of $\mathfrak{g}/\mathfrak{h}$. Using this data we can construct the vector bundle associated to the principal H-bundle via the adjoint representation, $G \times_{\mathrm{Ad}_H} \mathfrak{g}/\mathfrak{h}$. On the other hand, since G/H is a smooth manifold, we can also consider its tangent bundle $T[G/H] \to G/H$. It turns out that these two vector bundles are canonically isomorphic [Sha02].

$$\begin{tikzcd}
T[G/H] \arrow[rr, "\sim"] \arrow[dr] & & G \times_{\mathrm{Ad}_H} \mathfrak{g}/\mathfrak{h} \arrow[dl] \\
& G/H &
\end{tikzcd}$$

1.5.2 Cartan Geometry

In Minkowski spacetime we have a global Poincaré symmetry since there is no dependence on the points of the manifold. In gauge theory a powerful principle is to take a global symmetry and turn it into a local, or gauge, symmetry. A common example is the theory of electrically charged particles. Gauging the global U(1)-symmetry imposes the condition that an electromagnetic force appears. One can play the same game with the Poincaré symmetry of Minkowski spacetime. Gauging this global symmetry also causes a new force to appear; gravity.

One way to formally describe this process is through the reduction of a structure group. To every smooth manifold one can consider the principal frame bundle over it. Specifically this comes from the fact that every smooth manifold comes equipped with a metric, so one can define an $\mathrm{O}(n)$ principal bundle over the space. This $\mathrm{O}(n)$-bundle can be thought of as arising from the reduction of the structure group of a principal $\mathrm{GL}(n, \mathbb{R})$-bundle. The same idea works for the Lorentz group in indefinite signature. Using Klein geometries we can alternatively capture the reduction of structure group, as well as the connection data, in one object. Promoting to Cartan geometries is to consider a space modelled locally as a Kelin geometry, analagous to gauging the symmetry.

Definition 1.5.2. *A $H \hookrightarrow G$ **Cartan connection** over a manifold M consists of the following data.*

1. *A principal H-bundle $P_H \to M$.*
2. *A principal H-connection, $A \in \Omega^1(P_H; \mathfrak{g})$, such that:*
   (a) *There is a reduction of structure groups through $H \hookrightarrow G$;*
   (b) *and the 1-form, $A : TP_H \longrightarrow \mathfrak{g}$, induces an isomorphism $T_pP_H \cong \mathfrak{g}$, $\forall p \in P_H$.*

A space equipped with Cartan connection is a **Cartan geometry**. At each point there is a given fibre of the model Klein geometry. We can equip this space with a metric which differs from a Riemannian metric on a manifold in general. Here it characterises the size of the Klein geometry, and may contain some length scale relvant to the spacetime.

Definition 1.5.3. *A **metric** for a Klein geometry, (G, H), is a potentially degenerate Ad_H-invariant metric on $\mathfrak{g}/\mathfrak{h}$.*

If Riemannian curvature measures the deviation of a manifold from being flat, then Cartan geometry measures the deviation of a geometry from the model homogenous space.

Definition 1.5.4. *The **curvature** associated to a Cartan connection is defined by*

$$F = \mathrm{d}A + \frac{1}{2}[A \wedge A]. \tag{1.19}$$

We say that a Cartan geometry is **reductive** if the direct sum $\mathfrak{g} \cong \mathfrak{g}/\mathfrak{h} \oplus \mathfrak{h}$ is Ad_H-invariant. In our case we consider the pair $\mathrm{SO}(3,1) \hookrightarrow \mathrm{ISO}(3,1)$ as the 'model geometry'. Now the Lie algebra of the Poincaré group has a natural decomposition as,

$$\mathfrak{iso}(3,1) \cong \mathbb{R}^4 \oplus \mathfrak{so}(3,1).$$

Then the cartan connection can be decomposed as $A = \sigma + \omega$, where σ is a coframe field and ω is a principal $\mathrm{SO}(3,1)$-connection. Note that the vielbein is usually denoted by the letter, e, especially in the physics literature. Diagramatically we have:

$$\begin{tikzcd}
 & TP \arrow[d, "A"] \arrow[ddl, bend right, "\sigma"'] \arrow[ddr, bend left, "\omega"] & \\
 & \mathfrak{iso}(3,1) \arrow[dl] \arrow[dr] & \\
\mathbb{R}^4 & & \mathfrak{so}(3,1)
\end{tikzcd}$$

Assuming that the Lie algebra $\mathfrak{g}/\mathfrak{h}$ is abelian, then we can decompose the curvature as $F = \tau + \Omega$. Here Ω is the usual Riemann curvature 2-form and τ is the torsion.

$$\begin{tikzcd}
 & \bigwedge^2 TP \arrow[d, "F"] \arrow[ddl, bend right, "\tau"'] \arrow[ddr, bend left, "\Omega"] & \\
 & \mathfrak{iso}(3,1) \arrow[dl] \arrow[dr] & \\
\mathbb{R}^4 & & \mathfrak{so}(3,1)
\end{tikzcd}$$

Then Cartan's first structure equation is the vanishing of the torsion,

$$\tau^\alpha := \mathrm{d}\sigma^\alpha + \omega^\alpha_\beta \wedge \sigma^\beta. \tag{1.20}$$

Cartan's second structure equation gives a formula for the curvature in terms of only the connection,

$$\Omega^\alpha_\beta = \mathrm{d}\omega^\alpha_\beta + \omega^\alpha_\gamma \wedge \omega^\gamma_\beta. \tag{1.21}$$

The traditional Riemann curvature tensor is related to the curvature 2-form by,

$$\Omega^\alpha_\beta = \frac{1}{2} R^\alpha_{\beta\mu\nu} \mathrm{d}x^\mu \wedge \mathrm{d}x^\nu, \tag{1.22}$$

in a coordinate basis. To simplify computations we like to stay in the orthonormal basis defined by the vielbeins as opposed to the coordinate basis. In this basis we have some useful symmetry properties. The Lie algebra $\mathfrak{so}(3,1)$ satisfies [Frè13],

$$\omega \in \mathfrak{so}(3,1) \iff \omega^T \eta + \eta\omega = 0,$$

where η is the Minkowski metric. In a basis this can be written as $\omega^\gamma_\alpha \eta_{\beta\gamma} + \omega^\gamma_\beta \eta_{\alpha\gamma} = 0$. If we make the definition, $\omega_{\alpha\beta} := \eta_{\alpha\gamma}\omega^\gamma_\beta$, then this relation is often written,

$$\omega_{\alpha\beta} = -\omega_{\beta\alpha}. \tag{1.23}$$

Since we have a Lorentzian signature ω^α_β is not antisymmetric as in the Riemannian case. It picks up minus signs when swapping 'spatial' indices, but is symmetric under the interchange of the time and space indices. Note that in particular, the spin connection has $d^2(d-1)/2$ independant components, but the Christoffel symbols have $d^2(d+1)/2$. The fact there are less independant components is one of the reasons why this formalism is easier to work with for computations. In practice one starts with the coframe field σ^α_μ, which is obtained from (and equivalent to) the metric ansatz. Then we compute derivatives and write all expressions in the orthonormal frame. Using Cartan's structure equations we can deduce the forms of the connection $\omega^\alpha_{\mu\beta}$ and the curvature form $\Omega^\alpha_{\beta\mu\nu}$. Working with the vielbein field σ^α_μ and connection form $\omega^\alpha_{\mu\beta}$ is equivalent to working with a metric. The vilebein encodes the Lorentzian metric through,

$$g = -\sigma^0 \otimes \sigma^0 + \sum_{i=1}^{3} \sigma^i \otimes \sigma^i. \tag{1.24}$$

This is usually written

$$g_{\mu\nu} = \sigma^\alpha_\mu \sigma^\beta_\nu \eta_{\alpha\beta}, \tag{1.25}$$

in local coordinates, which is common in the physics literature. It's also common to treat the vielbein field (and their dual frame field) as an object which 'converts' spacetime indices to internal 'flat' indices. For example, a four-vector could be expressed as

$$A^\alpha = \sigma^\alpha_\mu A^\mu,$$

in a local orthonormal frame. Of course, this is just a change of basis applied to each ∂_μ in the tensor, which also works for the dual basis $\mathrm{d}x^\mu$ and frame fields. What is sometimes not emphasised in the physics literature is that these fields may not be well-defined **globally**. Indeed, it would be true if the manifold admits a **framing**, but this need not be a requirement for an arbitrary Lorentzian manifold. Globally hyperbolic spacetimes, discussed in Chapter 4, do actually enjoy this property howerver.

The above formalism is extremely insightful, but there's no new physics apparent yet, so what's the point? Beyond a better understanding of the theory, there are at least a few good reasons. The first is that it enables us to perform computations much easier, taking advantage of antisymmetry and differential forms. Another is that the universe contains fermions. We won't describe the details here, but the metric tensor alone isn't enough to describe spinor fields. In order to have a consistent unified theory, one must inevitably pass to the vielbein formalism to account for fermion fields in curved spacetime—[REF]—. This is especially true for supergravity and supersymmetric string theories, in which generalisations of the vielbein play important role. Even in classical gravity we can use Cartan geometry to decribe spacetimes which are not locally Minkowski. For example de Sitter space is described by the Klein geometry $\mathrm{SO}(4,1)/\mathrm{SO}(3,1)$. Then it might make sense to consider a spacetime which is locally modelled on this geometry as opposed to flat Minkowski space. This approach is useful in MacDowell-Mansouri gravity [Wis10]. For the moment, cosmological aspects of GR won't be discussed in these notes. One can assume that, when a vielbein field is introduced, we are working with the Cartan geometry modelled on $\mathrm{ISO}(3,1)/\mathrm{SO}(3,1)$.

Chapter 2

Schwarzschild Spacetime

2.1 Derivation of the Metric

The Schwarzchild solution is a static and spherically symmetric metric aiming to model the spacetime around a massive object. A derivation can be found in [Str13], we will follow this and also include explicit computations for all quantities. By definition, a static spacetime admits a timelike killing vector field, and we can decompose the manifold as a product

$$M = \mathbb{R}_t \times \Sigma,$$

for some spatial manifold Σ. In particular, the metric has the form

$$g = -f^2 \mathrm{d}t \otimes \mathrm{d}t + h, \tag{2.1}$$

where $f \in C^\infty(\Sigma)$ and h is a Riemannian metric on Σ. It is a vacuum solution and therefore satisfies $R_{\mu\nu}$. A spacetime is spherically symmetric if there is an SO(3) action which induces an isometry on the metric. Furthermore, we require that the orbits of this action are spacelike hypersuraces. If the spacetime is also static, as in this case, then the action is trivial on $\mathbb{R}_t$. This works when we explicitly make the assumption the timelike vector field is unique, so that there is a distinguished time coordinate. If ∂_t is unique then the SO(3) action must leave it invariant. The orbits of an isomtric SO(3) action are homogenous Riemannian manifolds of two dimensions. In this case it must have constant curvature, which implies that it is geodsically complete. Furthermore, this means it must be simply-connected.

Recall that the standard simply-connected surface is either the 2-sphere, the flat Euclidean plane, or the hyperbloic plane depending on the sign of the curvature. There is a known result that there exists a covering of our orbits by the standard surface such that the covering map is a local isometry. Since SO(3) is a subgroup of the isometry group of the orbit space, then the curvature must be strictly positive. From this we deduce the orbit space has to be the 2-sphere, S^2. Now we have a foliation of the spatial manifold as $\mathbb{R}_r \times S^2$, where r is the radial coordinate. This is **defined** through the area of the orbit sphere at that point, $A = 4\pi r^2$. Now we can write an ansatz for the metric.

$$g = -e^{2a(r)} \mathrm{d}t \otimes \mathrm{d}t + e^{2b(r)} \mathrm{d}r \otimes \mathrm{d}r + r^2 \mathrm{d}\theta \otimes \mathrm{d}\theta + r^2 \sin^2(\theta)\, \mathrm{d}\varphi \otimes \mathrm{d}\varphi \tag{2.2}$$

The spherical part of the metric is the standard round metric for a two-sphere of radius r. Note that we use exponentials since the functions have to be positive, we demand that $\mathrm{d}t \neq 0 \neq \mathrm{d}r$. The functions $a(r)$ and $b(r)$ do not depend on the other variables by the symmetry conditions. We will also impose that these functions satisfy **asymptotic flatness**, so that $r \to \infty \implies g \to \eta$. This is physically justified since we expect the gravitational field to taper off to zero at large distances. We won't use this until near the end of the derivation, but this is enforced by demanding that $a(r) \to 0$ as $r \to \infty$ since $e^a \to 1$; and similarly for $b(r)$. Furthermore, we include a factor of two for convenience in computation. Now we need to compute the Einstein tensor for the spacetime. We can read the coframes directly from the metric, $\sigma^\alpha = \sigma^\alpha_\mu \mathrm{d}x^\mu$ as:

$$\sigma^0 = e^{a(r)}\mathrm{d}t; \quad \sigma^1 = e^{b(r)}\mathrm{d}r; \quad \sigma^2 = r\mathrm{d}\theta; \quad \sigma^3 = r\sin\theta\mathrm{d}\phi.$$

Taking derivatives we obtain:

$$\mathrm{d}\sigma^0 = a'e^{-b}\sigma^1 \wedge \sigma^0; \quad \mathrm{d}\sigma^1 = 0;$$
$$\mathrm{d}\sigma^2 = \frac{e^{-b}}{r}\sigma^1 \wedge \sigma^2; \quad \mathrm{d}\sigma^3 = \frac{1}{r}\left(e^{-b}\sigma^1 \wedge \sigma^3 + \cot\theta\sigma^2 \wedge \sigma^3\right).$$

Now we use Cartan's first structure equation, $\mathrm{d}\sigma^\alpha = -\omega^\alpha_\beta \wedge \sigma^\beta$, to determine the connection form.

$$\mathrm{d}\sigma^0 = -\omega^0_1 \wedge \sigma^1 \implies \omega^0_1 = a'e^{-b}\sigma^0 = \omega^1_0$$
$$\mathrm{d}\sigma^2 = -\omega^2_1 \wedge \sigma^1 \implies \omega^2_1 = \frac{e^{-b}}{r}\sigma^2 = -\omega^1_2$$
$$\mathrm{d}\sigma^3 = -\omega^3_1 \wedge \sigma^1 - \omega^3_2 \wedge \sigma^2 \implies \omega^3_1 = \frac{e^{-b}}{r}\sigma^3 = -\omega^1_3;\ \omega^3_2 = \frac{\cot\theta}{r}\sigma^3 = -\omega^2_3$$

The curvature 2-form is computed using Cartan's second structure equation, $\Omega^\alpha_\beta = \mathrm{d}\omega^\alpha_\beta + \omega^\alpha_\gamma \wedge \omega^\gamma_\beta$.

$$\begin{aligned}\Omega^0_1 &= \mathrm{d}\omega^0_1 + \omega^0_\gamma \wedge \omega^\gamma_1 = \mathrm{d}\left(a'e^{-b}\right) \wedge \sigma^0 + a'e^{-b}\mathrm{d}\sigma^0 \\ &= \left(a''e^{-b} - a'b'e^{-b}\right) \wedge \sigma^0 + \left(a'\right)^2 e^{-b}\sigma^1 \wedge \sigma^0 \\ &\implies \Omega^0_1 = -e^{-2b}\left(a'' - a'b + \left(a'\right)^2\right)\sigma^0 \wedge \sigma^1\end{aligned}$$

$$\begin{aligned}\Omega^0_2 &= -\omega^0_1 \wedge \omega^1_2 = -a'e^{-b}\sigma^0 \wedge \frac{1}{r}e^{-b}\sigma^2 \\ &\implies \Omega^0_2 = \frac{a'}{r}e^{-2b}\sigma^0 \wedge \sigma^2\end{aligned}$$

$$\begin{aligned}\Omega^0_3 &= \omega^0_\gamma \wedge \omega^\gamma_3 = -a'e^{-b}\sigma^0 \wedge \frac{1}{r}e^{-b}\sigma^3 \\ &\implies \Omega^0_3 = -\frac{a'}{r}e^{-2b}\sigma^0 \wedge \sigma^3\end{aligned}$$

$$\Omega_2^1 = -\mathrm{d}\left(\frac{e^{-b}}{r}\right) \wedge \sigma^2 - \frac{e^{-b}}{r}\mathrm{d}\sigma^2$$
$$= \left(\frac{e^{-b}}{r^2} + \frac{b'}{r}e^{-b}\right)\mathrm{d}r \wedge \sigma^2 - \frac{e^{-2b}}{r}\left(\frac{1}{r} + b'\right)\sigma^1 \wedge \sigma^2 - \frac{e^{-2b}}{r^2}\sigma^1 \wedge \sigma^2$$
$$\implies \Omega_2^1 = \frac{b'}{r}e^{-2b}\sigma^1 \wedge \sigma^2$$

$$\Omega_3^2 = -\mathrm{d}\left(\frac{\cot\theta}{r}\right) \wedge \mathrm{d}\sigma^3 - \frac{\cot\theta}{r}\mathrm{d}\sigma^3 - \frac{e^{-2b}}{r^2}\sigma^2 \wedge \sigma^3$$
$$= \frac{1}{r^2}\cot\theta \mathrm{d}r \wedge \sigma^3 + \frac{1}{r\sin^2\theta}\mathrm{d}\theta \wedge \sigma^3 - \frac{\cot\theta}{r^2}\left(e^{-b}\sigma^1 \wedge \sigma^3 + \cot\theta\sigma^2 \wedge \sigma^3\right) - \frac{e^{-2b}}{r^2}\sigma^2 \wedge \sigma^3$$
$$\implies \Omega_3^2 = \frac{1 - e^{2b}}{r^2}\sigma^2 \wedge \sigma^3$$

Next we can compute the components of the Ricci tensor using $R_{\alpha\beta} = \Omega_\alpha^\gamma(\pi_\gamma, \pi_\beta)$, where $\{\pi_\alpha\}_{\alpha=0}^3$ is the dual basis to the coframes. For convenience write $A = -e^{-2b}\left(a'' - a'b + (a')^2\right)$.

$$R_{00} = \Omega_0^\gamma(\pi_\gamma, \pi_0) = -A + \frac{2a'}{r}e^{-2b}$$
$$R_{11} = \Omega_1^\gamma(\pi_\gamma, \pi_1) = A + \frac{2b'}{r}e^{-2b}$$
$$R_{22} = \Omega_2^\gamma(\pi_\gamma, \pi_2) = \frac{e^{-2b}}{r}(b' - a') + \frac{1}{r^2} - \frac{e^{-2b}}{r^2} = R_{33}$$

The Ricci scalar is then just the contraction with the Minkowski metric.

$$R = 2A + 4\frac{e^{-2b}}{r}(b' - a') + \frac{2}{r^2} - \frac{2e^{-2b}}{r^2}$$

Finally, the Einstein tensor components, $G_{\alpha\beta} = R_{\alpha\beta} - \frac{1}{2}Rg_{\alpha\beta}$, are as follows:

$$G_{00} = \frac{1}{r^2} - e^{-2b}\left(\frac{1}{r^2} - \frac{2b'}{r}\right)$$
$$G_{11} = -\frac{1}{r^2} + e^{-2b}\left(\frac{1}{r^2} + \frac{2a'}{r}\right)$$
$$G_{22} = e^{-2b}\left((a')^2 - a'b' + a'' + \frac{a' - b'}{r}\right) = G_{33}$$

Now we only have two functions to solve for, namely $a(r)$ and $b(r)$, using $G_{\alpha\beta} = 0$. Note that

$$G_{00} + G_{11} = \frac{2(b' + a')}{r} = 0 \implies b + a = C, \tag{2.3}$$

but we imposed asymptotic flatness meaning the integration constant C has to be zero to keep $a, b \to 0$. This means we can eliminate $a = -b$ and look at G_{00}.

$$\frac{1}{r^2} - e^{-2b}\left(\frac{1}{r^2} - \frac{2b'}{r}\right) = 0 \tag{2.4}$$

Multiplying through by r^2 we get

$$1 = e^{-2b} - 2b' r e^{-2b}, \tag{2.5}$$

where we can recognise the right-hand side as $(re^{-2b})'$. Now we can integrate both sides and pick up a constant $2m$,

$$r = re^{-2b} + 2m \implies e^{-2b} = 1 - \frac{2m}{r},$$

leaving us finally with the full metric

$$g = -\left(1 - \frac{2m}{r}\right) \mathrm{d}t \otimes \mathrm{d}t + \left(1 - \frac{2m}{r}\right)^{-1} \mathrm{d}r \otimes \mathrm{d}r + r^2 \mathrm{d}\theta \otimes \mathrm{d}\theta + r^2 \sin^2\theta \mathrm{d}\phi \otimes \mathrm{d}\phi. \tag{2.6}$$

One can show that m should be equal to GM/c^2, where G is Newton's gravitational constant and M has the interpretation as the mass of the central body. The reason is that we expect GR to reduce to Newtonian gravity in a sufficient limiting scenario. This is called the weak field approximation. More specifically, we consider only non-relativistic velocities and suppose that the gravitational field is weak, $g \approx \eta + h$, with $|h| \ll 1$. Then the time component of the metric satisfies

$$g_{00} \approx -\left(1 + \frac{2\Phi}{c^2}\right),$$

where Φ is the Newtonian gravitational potential obeying Poisson's equation

$$\nabla^2 \Phi = 4\pi G \rho,$$

for some mass distribution ρ.

2.2 Chart Problems

Importantly, there are two singularities appearing the above metric. The first of which occurs at the so-called event horizon $r = 2m$.

$$r \to 2m \implies g_{tt} \to 0,\ g_{rr} \to \infty$$

The second singularity occurs when $r = 0$.

$$r \to 0 \implies g_{tt} \to \infty,\ g_{rr} \to 0$$

Upon close analysis it's revealed that the first is actually just a coordinate singularity, meaning our choice of chart is at fault here. In particular there exists a change of coordinates such that the metric is well-behaved at the event horizon, which will be discussed momentarily. On the other hand, the singularity at $r = 0$ is very real, no matter what chart we use this is a problematic point in spacetime, and is often thought to be excised. Singularities are tricky since it is in general hard to tell when a choice of chart is at fault. Scalar curvature invariants are scalar quantities constructed from the Riemann curvature tensor. Since these are independent of the choice of chart, one can deduce the existence of a true singularity if they diverge. We can compute the *Kretschmann scalar* as

$$K := R_{\alpha\beta\gamma\delta} R^{\alpha\beta\gamma\delta} = 48 \frac{m^2}{r^6}.$$

It is evident that this function diverges for $r \to 0$. On the other hand $r = 2m$ is not problematic.

We must modify our definition of chart to account for this, in particular the image of the r coordinate map should be:

$$\mathrm{Im}(x^r) = (0, 2m) \cup (2m, \infty) \subset \mathbb{R}$$

Now, just because our chart doesn't see the points at $r = 2m$, it doesn't mean that this is not a part of the spacetime. We can extend the chart to cover the whole manifold, excluding the singularity, using a better choice of atlas.

2.2.1 Eddington-Finkelstein Coordinates

The quickest fix to the problem is to try to 'cancel out' the divergence at the event horizon with a counter term. We would like to search for a new function $f(r)$ such that

$$\frac{\mathrm{d}}{\mathrm{d}r} f(r) = \left(1 - \frac{2m}{r}\right)^{-1}. \tag{2.7}$$

This is solved by:

$$r_* \coloneqq f(r) = r + 2m \ln |r - 2m| \tag{2.8}$$

Now one can define a new **tortoise coordinate** $v \coloneqq t + r_*$ so that the metric takes the form:

$$\tilde{g} = -\left(1 - \frac{2m}{r}\right) \mathrm{d}v \otimes \mathrm{d}v + \mathrm{d}v \otimes \mathrm{d}r + \mathrm{d}r \otimes \mathrm{d}v + r^2 \mathrm{d}\theta \otimes \mathrm{d}\theta + r^2 \sin^2\theta \mathrm{d}\phi \otimes \mathrm{d}\phi \tag{2.9}$$

Computing the determinant of the metric we have

$$\det \tilde{g} = -\frac{\sin^2\theta}{r^4}, \tag{2.10}$$

which is evidently not singular at the event horizon. Moroever, the components $\tilde{g}_{\mu\nu}$ are all smooth functions, meaning that $\tilde{g}$ is now a well-defined metric on a larger spacetime, $\tilde{M}$, which includes the points

$$\{p \in \tilde{M} : x^v(p) \in \mathbb{R} \quad \& \quad x^r(p) \in (0, \infty)\}.$$

—[distinguish outgoing + ingoing]—In particular, we can now extend the original Schwarzschild spacetime (M, g) to a manifold which includes the event horizon $x^v(p) = 2m$ with the new metric $(\tilde{M}, \tilde{g})$. This is called an **analytic extension**—[define?]—.

The name 'event horizon' is motivated by the fact that information cannot escape a black hole. Note that information travels along future directed timelike/null curves in spacetime, or future directed causal curves. Consider such a timelike curve parameterised in these coordinates as $x^\mu(s) = (v(s), r(s), \theta(s), \phi(s))$. Then, if the curve is timelike

$$\tilde{g}_{\mu\nu} \dot{x}^\mu \dot{x}^\nu < 0 \implies \left(1 - \frac{2m}{r}\right) \dot{v}^2 + 2\dot{v}\dot{r} + \dot{r}^2(\dot{\theta} + \sin^2\theta \dot{\phi}^2) < 0,$$

which also implies

$$\dot{v}\left[2\dot{r} - \dot{v}\left(1 - \frac{2m}{r}\right)\right] < 0.$$

The velocity $\dot{v}$ does not change in sign. Here a choice of time-orientation is invoked. The standard convention is to take $\dot{v} > 0$, which implies

$$2\dot{r} - \dot{v}\left(1 - \frac{2m}{r}\right) < 0.$$

Now restricting to the region $\{p \in \tilde{M} : x^r(p) \leq 2m\}$ we notice that

$$-\dot{v}\left(1 - \frac{2m}{r}\right) \geq 0$$

which means that $\dot{r} < 0$ to preserve the inequality. So, along timelike curves, the radial coordinate must always decrease. In particular, massive particles can only cross the hypersurface $\{r = 2m\}$ when approaching from $\{r > 2m\}$. A similar argument also holds for lightlike curves as well, so the region contained within the event horizon is causally disconnected from the rest of spactetime.

2.2.2 Kruskal-Szekeres Spacetime

There are other atalses one can use to describe Schwarzschild spacetime. A particularly important choice is constructed using the following change of coordinates:

$$u(t,r) = \sqrt{\frac{r}{2m} - 1}e^{r/4m}\cosh\left(\frac{t}{4m}\right) \tag{2.11}$$

$$v(t,r) = \sqrt{\frac{r}{2m} - 1}e^{r/4m}\sinh\left(\frac{t}{4m}\right) \tag{2.12}$$

These are the **Kruskal** transformations. It's not immediately clear where these come from, but they can actually be derived from physical principles, as is done in [Str13]. Nothing special happens with the θ and φ coordinates so we can ignore those. Analysis of the transformed coordinates shows that u and v are in fact well-defined at $r = 2m$ and there is no longer a singularity. Once again, we can now consider the metric in the new coordinates with the points at $r = 2m$ included

$$g = \frac{32m^3}{r}e^{-r/2m}(\mathrm{d}v \otimes \mathrm{d}v - \mathrm{d}u \otimes \mathrm{d}u) + r^2(\mathrm{d}\theta \otimes \mathrm{d}\theta + \sin^2\theta \mathrm{d}\varphi \otimes \mathrm{d}\varphi). \tag{2.13}$$

—[Discuss maximal analytic extensions, Synge and Fronsdal?]—. The previous discussion highlights that singularities in general relativity are a delicate subject. They will be discussed more in Chapter 4.

2.3 Geodesics in Schwarzschild Spacetime

We will first use the Lagrangian description, which is equivalent to the Hamiltonian description in Chapter 1. As usual, we will make the abuse of notation $x^\mu(\tau) := (x^\mu \circ \gamma)(\tau)$ for the coordinates along a curve in spacetime, $\gamma : \mathbb{R} \to M$. The Lagrangian is then,

$$2\mathcal{L} = g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu = -1, \tag{2.14}$$

where the dot denotes differentiation with respect to proper time. Beacuse there is spherical symmetry, geodesics are confined to stay on a fixed plane. Without loss of generality we can fix this plane to be $\theta = \pi/2$ to simplify the setup. We then obtain the Lagrangian from the metric as,

$$2\mathcal{L} = -\left(1 - \frac{2m}{r}\right)\dot{t}^2 + \left(1 - \frac{2m}{r}\right)^{-1}\dot{r}^2 + r^2\dot{\varphi}^2. \tag{2.15}$$

Recall that symmetries of the metric, killing vectors, correspond to conserved quantities along geodesic flow. So, before even writing the geodesic equations, we have by definition two conserved quantities:

$$\begin{aligned} \mathcal{L}_{\partial_t} g = 0 &\implies -\left(1 - \frac{2m}{r}\right)\dot{t}^2 := -E \\ \mathcal{L}_{\partial_\varphi} g = 0 &\implies r^2\dot{\varphi} := L \end{aligned}$$

These correspond to an energy and axial angular momentum of the particle. Now we substitute the energy back into eq. (2.15) to obtain,

$$\dot{r}^2 + \left(1 - \frac{2m}{r}\right)\left(1 + \frac{L^2}{r^2}\right) = E^2. \tag{2.16}$$

Note also that we can express $\dot{r}$ in terms of $r' := \frac{\mathrm{d}r}{\mathrm{d}\varphi}$ with the chain rule, $\dot{r} = r'\dot{\varphi}$. Then substiuting the formula for angular momentum we have,

$$\frac{\mathrm{d}r}{\mathrm{d}\tau} = \frac{L}{r^2}\frac{\mathrm{d}r}{\mathrm{d}\varphi},$$

which can be substituted back into $2\mathcal{L} = -1$ to get:

$$\left(\frac{\mathrm{d}r}{\mathrm{d}\varphi}\right)^2 \frac{L^2}{r^4} + \left(1 + \frac{2m}{r}\right)\left(1 - \frac{L^2}{m}\right) = E^2. \tag{2.17}$$

This can be solved exactly by using the transformation $u = 1/r$, but we won't do this here. The upshot is that, for massive bounded particles, the geodesics are ellipses when projected to a three dimesnional spatial slice. This agrees with what's expected from Newtonian gravity. The difference, however, is that the ellipses exhibts **precession** over time. One of the first predictions of general relativity was in fact the famous 'anomalous perihelion of Mercury', which can be derived with remarkable accuracy using Einstein's theory.

In preparation for the Kerr metric, it will be useful exercise to work with the Hamiltonian formalism and derive the equations of motion again. —[DO THIS]—

Chapter 3

Kerr Spacetime

3.1 Metric

Kerr spacetime is again a vacuum solution to the Einstein's field equations. It models the field generated by rotating matter, and like the Schwarzschild case it contains a singularity. In Schwarzschild spacetime we had a spherical symmetry, in this case we loosen this to an axial one. The metric will also be static, hence there will be something similar to a timelike killing field. Note that this is slightly strange considering the mass is spinning constantly (and hence not 'static'), but we're assuming that the angular momentum is fixed for all time.

There's not really a quick and conceptually intuitive way to derive this metric, unlike in for Schwarzschild spacetime. There is a derivation in Straumann [Str13] which serves the purpose of being more general, in that it can be extended to account for rotating stars as well, as opposed to just the vacuum surrounding a rotating black hole solution. A quicker but less conceptual derivation can be done by 'analytically rotating' into complex coordinates covered in [New73]. One can also 'brute force' the calculation by considering an ellipsoidal ansatz [NP13], which is more grounded but perhaps tedious in terms of computation—[do and include?]—. All of the above approaches result in the same metric regardless, albeit in many incarnations of coordinates. We will focus on analysing the following expression in Boyer-Lindquist coordinates —[DEFINE]—.

$$ds^2 = -\frac{\Delta}{\rho}(dt - a\sin^2\theta d\phi)^2 + \frac{\sin^2\theta}{\rho^2}((r^2+a^2)d\phi - adt)^2 + \frac{\rho}{\Delta}dr^2 + \rho^2 d\theta^2 \tag{3.1}$$

Here, $\Delta = r^2 - 2mr + a^2$, $\rho^2 = r^2 + a^2\cos^2\theta$, and $a = J/m$ is the **Kerr parameter**. The

3.2 Analysing the Geometry of Kerr Spacetime

For this section we will closely follow relevant parts of [Vis08], which is an excellent introductory overview of aspects Kerr spacetime. Explicitly writing out all terms in the line element will make

parts of the analysis in this section simpler.

$$
\begin{aligned}
g = &-\left(1 - \frac{2mr}{r^2 + a^2\cos^2\theta}\right) \mathrm{d}t \otimes \mathrm{d}t \\
&- \frac{2mra\sin^2\theta}{r^2 + a^2\cos^2\theta}(\mathrm{d}t \otimes \mathrm{d}\phi + \mathrm{d}\phi \otimes \mathrm{d}t) \\
&+ \left(\frac{r^2 + a^2\cos^2\theta}{r^2 - 2mr + a^2}\right) \mathrm{d}r \otimes \mathrm{d}r + (r^2 + a^2\cos^2\theta)\mathrm{d}\theta \otimes \mathrm{d}\theta \\
&+ \left(r^2 + a^2 + \frac{2mra^2\sin^2\theta}{r^2 + a^2\cos^2\theta}\right) \sin^2\theta \mathrm{d}\phi \otimes \mathrm{d}\phi
\end{aligned}
\tag{3.2}
$$

Kerr geometry is significantly more involved that Schwarzschild case, as one may intuitively guess from the metric. To summarise, there are several 'subspaces' of importance. —[DRAW PICTURE]—

Remark 3.2.1. *Kerr spacetime has two event horizons, two 'ergosurfaces', and one singularity. Going inwards by decreasing r, one encounters, with possible intersection:*

1. *An **outer ergosurface** $r_{\mathcal{E}}^{+} = m + \sqrt{m^2 - a^2\cos^2\theta}$;*
2. *An **outer event horizon** $r_{+} = m + \sqrt{m^2 - a^2}$;*
3. *An **inner event horizon** $r_{-} = m - \sqrt{m^2 - a^2}$;*
4. *An **inner ergosurface** $r_{\mathcal{E}}^{-} = m - \sqrt{m^2 - a^2\cos^2\theta}$;*
5. *A **ringularity** at $r = 0$ & $z = 0$.*

3.2.1 Horizons

Let us start by studying the horizons. It is clear from inspecting eq. (3.2) that the g_{rr} term diverges if

$$r^2 - 2mr + a^2 = 0, \tag{3.3}$$

which is easily solved to obtain:

$$r_{\pm} = m \pm \sqrt{m^2 - a^2} \tag{3.4}$$

Now these are not true singularites. Indeed, like in the Schwarzschild case, one can check that the curvature invariant $R^{\mu\nu\rho\sigma}R_{\mu\nu\rho\sigma}$ is finite. Now fix the r coordinate to a value $r_* \in \mathbb{R}_{>0}$ and let the other coodinates vary. Denote the resulting codimension-1 hypersurface by $\sigma : \Sigma \hookrightarrow M$.

$$\Sigma := \{p \in M : x^r(p) = r_* \in \mathbb{R}_{>0}\} \tag{3.5}$$

Then one can consider the induced metric $\sigma^\star g$ by setting $\mathrm{d}r = 0$, giving the associated line element:

$$ds_{\Sigma}^2 = -dt^2 + (r_*^2 + a^2\cos^2\theta)d\theta^2 + (r_*^2 + a^2)\sin^2\theta d\phi^2 + \frac{2m}{r_*}\frac{(dt - a\sin^2\theta d\phi)^2}{1 + a^2\cos^2\theta} \tag{3.6}$$

Now we can compute the determinant of the induced metric as:

$$\det(\sigma^\star g) = -\sin^2\theta(r_*^2 + a^2\cos^2\theta)(r_*^2 - 2mr_* + a^2) \tag{3.7}$$

It's clear that $\sin^2\theta(r_*^2+a^2\cos^2\theta)$ is always positive, and so the sign of the determinant is determined entirely by $f(r_*) = r_*^2 - 2mr_* + a^2$. For values outside both horizons, $r_* > r_+$ or $r_* < r_-$, we have $f(r_*) > 0 \implies \det\sigma^\star g < 0$. This is expected for a $(2+1)$-dimensional Lorentzian manifold. On the other hand for the region between the horizons, $r_* \in (r_-, r_+)$, we have $f(r_*) < 0 \implies \det\sigma^\star g > 0$. In this case the metric has a Euclidean signature, and hence it is hard to make sense of a 'time direction'. The transition point occurs at the horizons, $r_* = r_\pm$, which are the roots of $f(r_*)$. In particular the matrix of the induced metric is singular here $\det\sigma^\star g = 0$. Note that does **not** imply there is a curvature singularity present, these are merely coodinate singularites analagous to the event horizon in Schwarzschild spacetime.

Denote the hypersurfaces of the inner and outer event horizons by $\Sigma_\pm$. Since $\sigma^\star g$ is singular,

$$\forall p \in \Sigma_\pm\ \exists \ell \in T_p\Sigma_\pm : [\sigma^\star g]_{ij}\ell^i = 0,$$

moreover, it follows that $(\sigma^\star g)(\ell,\ell) = 0$. Now these vectors are only defined on the tangent space on the hypersurfaces. We can lift them to a tangent vector field on spacetime by $\ell^i \mapsto (\ell^t, 0, \ell^\theta, \ell^\phi)$. At the event horizons we have that $g_{\mu\nu}\ell^\mu\ell^\nu = 0$, so this ℓ is a null vector field tangent to a null geodesic which skims along the surface of a horizon. —[describe 'rotation' of horizons?]—.

Let us reduce by one more dimension by fixing the time coordinate to $t = 0$. Now to study just the horizons we also restrict our attention to $r_* = r_\pm$. Denote the resulting 2-dimensional surface by:

$$\Pi_\pm \coloneqq \{p \in M : x^r(p) = r_\pm \ \&\ x^t(p) = 0\} \tag{3.8}$$

One might suspect that these surfaces are spherical, like the horizon in the Schwarzschild case, however this isn't quite right. What is true is that they are topologically 2-spheres $\Pi_\pm \cong_{\text{Top}} S^2$, but the metric is quite different. More concretely they are not isometric to the 2-sphere with the round metric. To understand this we can again look at the induced metric $\pi^\star g$, where the inclusion is $\pi : \Pi_\pm \hookrightarrow M$.

$$\pi^\star g = (r_\pm^2 + a^2\cos^2\theta)\mathrm{d}\theta\otimes\mathrm{d}\theta + \frac{4m^2r_*^2}{r_*^2 + a^2\cos^2\theta}\sin^2\theta\mathrm{d}\phi\otimes\mathrm{d}\phi \tag{3.9}$$

The area of the surfaces can be computed [Vis08]:

$$A(\Pi_\pm) = 4\pi(r_\pm^2 + a^2) \tag{3.10}$$

The Ricci scalar is given by:

$$R_{\Pi_\pm}(\theta) = \frac{2(r_\pm^2 + a^2)(r_\pm^2 - 3a^2\cos^2\theta)}{(r_\pm^2 + a^2\cos^2\theta)} \tag{3.11}$$

We can study the bahaviour of this function as θ varies. For $\theta \to \pi/2$, the equator, we have:

$$R_{\Pi_\pm}(\pi/2) = \frac{2(r_\pm^2 + a^2)}{r_\pm^4} \tag{3.12}$$

In particular this is expression is always greater than zero. On the other hand, on the axis of symmetry $\theta = 0$ or $\theta = \pi$, eq. (3.11) becomes:

$$R_{\Pi_\pm}(\pi) = R_{\Pi_\pm}(0) = \frac{2(r_\pm^2 + a^2)(r_\pm^2 - 3a^2)}{(r_\pm^2 + a^2)^3} \tag{3.13}$$

Notice that near the poles the intrinsic curvature can actually become negative. Exactly on the axis for example, this occurs when $a^2 > r_\pm^2/3$. For the outer horizon r_+ this corresponds to $a > \sqrt{3}m/2$, even before the extremal case for $a = m$. More detail the intrinsic geometry of the horizons can be found in [Vis08].

3.2.2 Ergosurfaces

The ergosurfaces in Kerr spacetime can be chracterised in a coordinate free way. Denote the Killing vector associated to translation in time by $K_{[t]} \in \mathfrak{X}(M)$. This is the unique Killing vector field such that $g(K_{[t]}, K_{[t]}) \to -1$ asymptotically far away from the black hole, or at spatial infinity. The other Killing vector field associated to Kerr spacetime is the axial symmetry $K_{[\phi]} \in \mathfrak{X}(M)$. These are the only two Killing vector fields up to linear combinations of each other.

Definition 3.2.1. *In Kerr spacetime an* ***ergosurface*** *is where the Killing vector field associated to time translation becomes null.*

$$\mathcal{E} \coloneqq \{p \in M : g_p(K_{[t]}, K_{[t]}) = 0\} \tag{3.14}$$

—[also give invariant characterisation of horizons?]— Physically, the ergosurfaces can be understood also in coordinates. Consider an observer, γ, who is 'staying still' in Boyer-Lindquist coordinates, and so has no spacelike velocity:

$$\dot{\gamma} = \partial_t \tag{3.15}$$

If an observer is stationary then they are necessarily timelike, $g(\dot{\gamma}, \dot{\gamma}) < 0$, but in this case:

$$\begin{aligned} g(\partial_t, \partial_t) &= g_{tt} < 0 \\ \implies -\left(1 - \frac{2mr}{r^2 + a^2\cos^2\theta}\right) &< 0 \end{aligned}$$

Define the zeros of $g_{tt}(r)$ as:

$$r_\mathcal{E}^\pm \coloneqq m \pm \sqrt{m^2 - a^2\cos^2\theta} \tag{3.16}$$

So if the observer wishes to remain timelike, they must stay outside the region $r \in (r_\mathcal{E}^-, r_\mathcal{E}^+)$. In particular, it is impossible to remain stationary between the surfaces $r = r_\mathcal{E}^\pm$. We can now form a sequence of inequalities

$$r_\mathcal{E}^+ \geq r_+ \geq r_- \geq r_\mathcal{E}^-, \tag{3.17}$$

with the outer two being equalities only on the axis of rotation $\theta = 0, \pi$.

Similar to when discussing horizons, we can look at the intrinsic geometry of ergosurfaces. Fixing the x^r coordinate to $r = r_\mathcal{E}^\pm$ we denote the resuling codimension-1 hypersurfaces by $\varepsilon : \mathcal{E}_\pm \hookrightarrow M$.

$$\mathcal{E}_\pm \coloneqq \{p \in M : x^r(p) = r_\mathcal{E}^\pm\} \tag{3.18}$$

The induced metric is then:

$$\begin{aligned}\varepsilon^\star g = &- a\sin^2\theta(\mathrm{d}t\otimes\mathrm{d}\phi+\mathrm{d}\phi\otimes\mathrm{d}t)\\ &+\frac{2mr^3r_{\mathcal{E}}^{\pm}}{m^2-a^2\cos^2\theta}\mathrm{d}\theta\otimes\mathrm{d}\theta\\ &+2(mr_{\mathcal{E}}^{\pm}+a^2\sin^2\theta)\sin^2\theta\mathrm{d}\phi\otimes\mathrm{d}\phi\end{aligned} \tag{3.19}$$

We can play the same game yet again and look at the induced metric at a given intance of time $t=0$. Then there are associated 2-dimesional surfaces $\delta:\Delta_\pm\hookrightarrow M$ in spacetime denoted:

$$\Delta_\pm \coloneqq \{p\in M : x^r(p)=r_{\mathcal{E}}^{\pm}\ \&\ x^t(p)=0\} \tag{3.20}$$

—[exact results get complicated here, find and complete?]—

3.3 Integrability of Kerr Geodesic Flow

Kerr geodesic flow has three obvious constants of motion, two of which arise from two obvious symmetries of the metric. Since we have geodesic flow there is also the Hamiltonian itself, or the rest mass of the particle. Another conserved quantity arises from the axial symmetry generated by ∂_φ, implying that there is a conserved momentum $p_\varphi = g_{\mu\varphi}p^\mu$. Furthermore, there is a timelike killing field which induces a conserved p_t associated to a kind of energy. There is in fact another 'hidden symmetry' associated to the total phase space $T^\star M$. We say hidden since this is not a lift of a symmetry in coniguration space, like usual translations along spacetime coordinates for example. The existence of such a transformation is usually hard to deduce, especially in 8-dimensional curved phase space.

Instead, we can go the other way and infer the existence of a global conserved quantity. This is known as Carter's constant. Then, by Noether's theorem, this does correspond to a symmetry vector field in phase space. Though, for this discussion, the symmetry vector field itself won't be as important. Recall that the existence of four Poisson commuting and independant conserved quantities makes geodesic flow **integrable**. In particular, this means that we can solve the geodesic equation, at least by quadratures.

3.3.1 The Fourth Integral

—[Give exposition on geometric formulation of H-J + Killing tensors]—. The Hamilton-Jacobi equation is in general given by:

$$H\left(x^\mu,\frac{\partial S}{\partial x^\nu}\right)+\frac{\partial S}{\partial\lambda}=0 \tag{3.21}$$

One would like to solve for a complete integral $S=S(x^\mu,\lambda,\alpha_\nu)$ with four parameters α_ν. We impose the condition that:

$$\det\left[\frac{\partial^2 S}{\partial x^\mu\partial\alpha_\nu}\right]\neq 0 \tag{3.22}$$

The interpretation of S is that it is a generating function for a canonical transformation. Specifically, from the orignal system with variables (x^μ,p_μ) and Hamiltonian H to a new system with variables

(β^μ, α_μ) and Hamiltonian H'. The relationships are given by:

$$p_\mu = \frac{\partial S}{\partial x^\mu} \tag{3.23}$$

$$\beta^\mu = \frac{\partial S}{\partial \alpha_\mu} \tag{3.24}$$

$$H' = H + \frac{\partial S}{\partial \lambda} \tag{3.25}$$

If S is a solution then $H' = 0$, so α and β are constants of motion. One can then invert the equations to solve for $(x(\lambda)^\mu, p_\mu(\lambda))$. Since H is time-independent we have the seperation

$$S = \frac{1}{2}\mu^2\lambda + W(x^\mu), \tag{3.26}$$

and W satisfies the *reduced* Hamilton-Jacobi equation:

$$H\left(x^\mu, \frac{\partial W}{\partial x^\nu}\right) = \frac{1}{2}\mu^2 \tag{3.27}$$

In the case of geodesic flow this is just

$$g^{\mu\nu}\frac{\partial W}{\partial x^\mu}\frac{\partial W}{\partial x^\nu} + \mu^2 = 0. \tag{3.28}$$

Using $p_\mu = \partial_\mu S$ we also have

$$g^{\mu\nu}p_\mu p_\nu + \mu^2 = 0, \tag{3.29}$$

so μ is the rest mass of the particle. Now looking at the Kerr metric this becomes:

$$\frac{\Delta}{\rho^2}(\partial_r W)^2 + \frac{1}{\rho^2}(\partial_\theta W)^2 + \frac{\Delta - a^2\sin^2\theta}{\rho^2\Delta\sin^2\theta}(\partial_\phi W)^2 - \frac{4mar}{\rho^2\Delta}\partial_\phi W\partial_t W - \frac{\Sigma}{\rho^2\Delta}(\partial_t W)^2 + \mu^2 = 0 \tag{3.30}$$

Note that we have used the shorthands $\rho^2 = r^2 + a^2\cos^2\theta$, $\Delta = r^2 - 2mr + a^2$, and $\Sigma = r^2 + a^2 - a^2\sin^2\theta$.

This is independent of t and ϕ so we try the seperation ansatz

$$W = -Et + L_z\phi + S_r(r) + S_\theta(\theta), \tag{3.31}$$

where E and L_z are the conserved quantities assocated to the time translation and axial Killing vectors:

$$-E \coloneqq \langle K_{[t]}, p\rangle \quad \& \quad L_z \coloneqq \langle K_{[\phi]}, p\rangle \tag{3.32}$$

Inserting in the ansatz we obtain

$$\begin{aligned} &\Delta\left(\frac{\mathrm{d}S_r}{\mathrm{d}r}\right)^2 + \left(\frac{\mathrm{d}S_\theta}{\mathrm{d}\theta}\right)^2 + \frac{\Delta - a^2\sin^2}{\Delta\sin^2\theta}L_z^2 \\ &+\frac{4mar}{\Delta}L_zE - \frac{\Sigma}{\Delta}E^2 + \rho^2\mu^2 = 0 \end{aligned} \tag{3.33}$$

Now we can gather terms which depend on θ and those which depend on r:

$$\Delta\left(\frac{\mathrm{d}S_r}{\mathrm{d}r}\right)^2 + \frac{4mar}{\Delta}L_zE - \frac{(r^2+a^2)^2}{\Delta}E^2 - \frac{a^2}{\Delta}L_z^2 + r^2\mu^2$$
$$+$$
$$\left(\frac{\mathrm{d}S_\theta}{\mathrm{d}\theta}\right)^2 - \frac{L_z^2}{\sin^2\theta} + a^2E^2\sin^2\theta + a^2\mu^2\cos^2\theta = 0$$

Each of these terms must constant. Adding and subtracting $a^2E + L_z^2$ we can rearrange to obtain:

$$\left(\frac{\mathrm{d}S_\theta}{\mathrm{d}\theta}\right)^2 + \mu^2\cos^2\theta + a^2E^2(\sin^2\theta - 1) + \left(\frac{1}{\sin^2\theta} - 1\right)L_z^2 = C \tag{3.34}$$

$$\implies p_\theta^2 + \cos^2\theta\left(a^2(\mu^2 - E^2) + \frac{1}{\sin^2\theta}L_z^2\right) = C \tag{3.35}$$

We call the seperation constant the **Carter constant** after its discoverer Brandon Carter. Similarly, the r-dependent equation takes the form:

$$\Delta\left(\frac{\mathrm{d}S_r}{\mathrm{d}r}\right)^2 = \frac{1}{\Delta}\left[E(r^2+a^2) - aL_z\right]^2 - \left[C + (L_z - aE)^2 + \mu^2r^2\right] \tag{3.36}$$

For convenience, we define the following functions

$$R(r) \coloneqq \left[E(r^2+a^2) - aL_z\right]^2 - \Delta\left[C + (L_z - aE)^2 + \mu^2r^2\right] \tag{3.37}$$
$$\Theta(\theta) \coloneqq C - \cos^2\theta\left[a^2(\mu^2 - E^2) + \frac{1}{\sin^2\theta}L_z^2\right], \tag{3.38}$$

and finally the solution of the Hamilton-Jacobi equation is:

$$S = \frac{1}{2}\mu^2\lambda - Et + L_z\phi + \int^r \frac{1}{\Delta}\sqrt{R}\mathrm{d}r + \int^\theta \sqrt{\Theta}\mathrm{d}\theta \tag{3.39}$$

Using the four integrals, one can immediately obtain the following ordinary differential equations:

$$\rho^2\dot{\theta} = \sqrt{\Theta} \tag{3.40}$$
$$\rho^2\dot{r} = \sqrt{R} \tag{3.41}$$
$$\rho^2\dot{\phi} = -\left(aE - \frac{L_z}{\sin^2\theta}\right) + \frac{a}{\Delta}\left[E(r^2+a^2) - L_za\right] \tag{3.42}$$
$$\rho^2\dot{t} = -a\left(aE\sin^2\theta - L_z\right) + \frac{r^2+a^2}{\Delta}\left[E(r^2+a^2) - L_za\right] \tag{3.43}$$

Now $\theta(\lambda)$ and $r(\lambda)$ can be computed via integration, which can then be inserted into the final two equations to obtain the full trajectory via quadratures again.

Chapter 4

Causality & Topology

4.1 Obstructions to Spacetime

The main goal of this chapter will be to understand the causal nature of spacetime and cover Penrose's singularity theorem. We will mainly follow Witten's [Wit20], as well as Galloway's [Gal14] lectures. Note that we won't be proving every statement carefully, such proofs can be found in [O'N83] and references therein. By spacetime, we mean as defined in Definition 1.1.4, but we allow arbitrary $d \coloneqq \dim M \geq 2$, and may relax the orientation for the purpose of discussion. There are a lot of subtelties involved in determing whether a manifold is/can be a spacetime. For example, not all Lorentzian manifolds are time-orientable, even if the underlying manifold is orientable. Consider the manifold $\mathbb{R} \times [0, 2\pi)$ with opposite sides indentified, $\mathbb{R} \times S^1$. In this case the cylinder is of course orientable, but a time-orientation cannot always be chosen. Take the standard metric given by

$$g = -\mathrm{d}t \otimes \mathrm{d}t + \mathrm{d}\theta \otimes \mathrm{d}\theta,$$

where $\theta \sim \theta + 2\pi$ is the standard coordinate on the circle. It's clear that the vector ∂_t defines a time-orientation since it's globally well-defined and nowhere vanishing. On the other hand consider the following counter-example.

Proposition 4.1.1. *The manifold $\mathbb{R} \times S^1$ is Lorentzian when equipped with the metric*

$$g = -\cos(\phi)\mathrm{d}x \otimes \mathrm{d}x + \sin(\phi)(\mathrm{d}x \otimes \mathrm{d}\phi + \mathrm{d}\phi \otimes \mathrm{d}x) + \cos(\phi)\mathrm{d}\phi \otimes \mathrm{d}\phi,$$

yet $(\mathbb{R} \times S^1, g)$ is not time-orientable.

Proof. To show that $\mathbb{R} \times S^1, g$ is Lorentzian we need to show that it can be put in diagonal form with eigenvalues $(-1, +1)$. Consider the following vector fields,

$$\sigma_0 \coloneqq \cos(\varphi)\partial_x - \sin(\varphi)\partial_\phi; \quad \sigma_1 \coloneqq \sin(\varphi)\partial_x + \cos(\varphi)\partial_\phi,$$

where $\varphi \coloneqq \phi/2$ for convenience. We would like to show that these define orthonormal frames for

the metric. First let's show that $\langle \sigma_0, \sigma_0 \rangle = -1$:

$$\begin{aligned} g(\sigma_0, \sigma_0) &= -\cos\phi \, \mathrm{d}x\big(\cos\varphi\partial_x - \sin\varphi\partial_\phi\big)^2 \\ &+ 2\sin\phi \, \mathrm{d}x\big(\cos\varphi\partial_x - \sin\varphi\partial_\phi\big) \otimes \mathrm{d}\phi\,(\cos\varphi\partial_x - \sin\varphi\partial_\phi) + \cos\phi \, \mathrm{d}\phi\big(\cos\varphi\partial_x - \sin\varphi\partial_\phi\big)^2 \\ &= -\cos\phi\cos^2\varphi - 2\sin\phi\cos\varphi\sin\varphi + \cos\phi\sin^2\varphi \\ &= -\cos\phi(\cos^2\varphi - \sin^2\varphi) - \sin^2\phi = -1 \end{aligned}$$

Similarly, for $\langle \sigma_1, \sigma_1 \rangle$:

$$\begin{aligned} g(\sigma_1, \sigma_1) &= -\cos\phi \, \mathrm{d}x\big(\sin\varphi\partial_x + \cos\varphi\partial_\phi\big)^2 \\ &+ 2\sin\phi \, \mathrm{d}x\big(\sin\varphi\partial_x + \cos\varphi\partial_\phi\big) \otimes \mathrm{d}\phi\,(\sin\varphi\partial_x + \cos\varphi\partial_\phi) + \cos\phi \, \mathrm{d}\phi\big(\sin\varphi\partial_x + \cos\varphi\partial_\phi\big)^2 \\ &= -\cos\phi\sin^2\varphi + 2\sin\phi\cos\varphi\sin\varphi + \cos\phi\cos^2\varphi \\ &= \cos\phi(\cos^2\varphi - \sin^2\varphi) + \sin^2\phi = 1 \end{aligned}$$

Finally, the cross term $\langle \sigma_0, \sigma_1 \rangle$:

$$\begin{aligned} g(\sigma_0, \sigma_1) = &-\cos\phi \, \mathrm{d}x\,(\cos\varphi\partial_x - \sin\varphi\partial_\phi) \otimes \mathrm{d}x\,(\sin\varphi\partial_x + \cos\varphi\partial_\phi) \\ &+ \sin\phi \, \mathrm{d}x\,(\cos\varphi\partial_x - \sin\varphi\partial_\phi) \otimes \mathrm{d}\phi\,(\sin\varphi\partial_x + \cos\varphi\partial_\phi) \\ &+ \sin\phi \, \mathrm{d}\phi\,(\cos\varphi\partial_x - \sin\varphi\partial_\phi) \otimes \mathrm{d}x\,(\sin\varphi\partial_x + \cos\varphi\partial_\phi) \\ &+ \cos\phi \, \mathrm{d}\phi\,(\cos\varphi\partial_x - \sin\varphi\partial_\phi) \otimes \mathrm{d}\phi\,(\sin\varphi\partial_x + \cos\varphi\partial_\phi) \\ = &-2\cos\phi\cos\varphi\sin\varphi + \sin\phi\,\big(\cos^2\varphi - \sin^2\varphi\big) = 0 \end{aligned}$$

So g is indeed a valid Lorentzian metric. Now suppose for contradiction that $(\mathbb{R} \times S^1, g)$ is time-orientable. Then there exists a nowehere vanishing global timelike section, $T \in \mathfrak{X}(M)$, such that $g(\sigma_0, T)|_p < 0$, where $x^\mu(p) = (0, 0)$. Now define a function, $f(s) \coloneqq g(X_0, T)(\gamma(s))$, where $\gamma : [0, 1) \to \mathbb{R} \times [0, 2\pi)$ is a smooth curve. We have that $f : [0, 1) \to \mathbb{R}$ is a smooth function since γ and g are both smooth. Furthermore, assume that the curve satisfies the initial condition $\gamma(0) = p$. We would like to show that $g(\sigma_0, T)|_\gamma < 0$ for all points on the curve. Suppose for contradiction that $\exists t \in [0, 1) : f(t) < 0$. Then $g(X_0, T)(\gamma(t)) = 0$. But we also know that $g(\sigma_0, \sigma_0) = -1$, so σ_0 is timelike. If $g(\sigma_0, T)|_{\gamma(t)} = 0$ then T must be spacelike since it is orthognal to the timelike vector σ_0. Then this is a contradiction since T was assumed to be timelike everywhere. Furthermore, we have $f(0) < 0$, so by continuity $f(s) < 0 \, \forall s \in [0, 1)$. Now consider a curve such that $\gamma(s) = (0, 2\pi s)$. We have that $\lim_{s\to 1} g(\sigma_0, T)(\gamma(s)) = -g(\sigma_0, T)(\gamma(0))$ since $\lim_{s\to 1} \sigma_0(0, 2\pi s) = -\partial_x = -\sigma_0(0, 0) = \partial_x$. Recall that $\mathbb{R} \times S^1$ is defined by identifying $\phi \sim \phi + 2\pi$, in particular $0 \sim 2\pi$. Now $f(0) = -\lim_{s\to 1} f(s) \sim -f(0)$ in the quotient, hence $f = 0$, a contradiction—[IRON THIS OUT]—. $\square$

There is a nice visualisation of the above exmaple. The frame fields precisely define a smooth rotation as ϕ varies along the horizontal of the strip $\mathbb{R} \times [0, 2\pi)$. The half-angle results in the roation approaching a flip along the horizontal as $\phi \to 2\pi$. When we pass to the quotient, the two vertical edges of the strip are identified. In this case a light cone is smoothly transported around the cylinder while rotating, until it is upside down when it comes back to its starting point. Hence, there is no way to 'pick' either the top or bottom half-cone unambiguously.

Time orientability is defined as a global property, since every Lorentzian manifold admits local timelike sections. There is already a topological obstruction to the existence of global nowhere

vanishing sections in the **Euler class** of the tangent bundle. —[REF]—. Howerver, even if there exists an orientable Lorentzian manifold with a nowehere vanishing section, there is a further geometric obstruction from the metric itself. So time-orientability is both a topological and geometric property of the space. On the other hand the potential for an arbitrary smooth manifold to *admit* a time-orientable Lorentzian metric is more relaxed.

Proposition 4.1.2. *Any smooth manifold M with a nowhere vanishing global section $X \in \mathfrak{X}(M)$ admits a time-orientable Lorentzian metric.*

Proof. Since M is a smooth manifold it admits a Riemannian metric h, with musical isomorphism $\flat : TM \longrightarrow T^\star M$. Then define a new metric by

$$g_X \coloneqq h - \frac{2}{h(X,X)} X^\flat \otimes X^\flat. \tag{4.1}$$

One can show—[FINISH PROOF]— that this new metric is indeed Lorentzian and time-orientable. □

There is a nice packaging of these statements into the following theorem [MS06].

Theorem 4.1.1. *Suppose M is a smooth and connected manifold. The following statements are equivalent:*

1. *M admits a Lorentzian metric.*
2. *M admits a time-orientable Lorentzian metric.*
3. *M admits a smooth nowhere vanishing global section $X \in \mathfrak{X}(M)$.*
4. *M is either non-compact or has vanishing Euler characteristic $\chi(M) = 0$.*

—[a lot more to say here]—

4.2 Causal Relations

Recall that curves in spacetime can be characterised into three groups.

Definition 4.2.1. *A curve $\gamma : (a,b) \longrightarrow M$ is:*

1. ***timelike** if $g(\dot\gamma, \dot\gamma) < 0$;*
2. ***null** if $g(\dot\gamma, \dot\gamma) = 0$;*
3. ***spacelike** if $g(\dot\gamma, \dot\gamma) > 0$;*

for all points in the image of γ.

We say that a curve is causal if it is either timelike or null. Intuitively, matter particles must travel along timelike curves and photons along null curves. A curve γ is **future (past) directed** if $\dot\gamma$ is future (past) directed everywhere. A future directed causal curve $\gamma : (a,b) \longrightarrow M$ is **future inextendible** in M if $\lim_{\tau \to b^-} \gamma(\tau)$ does not exist. Similarly one has a notion of a curve being past inextendible. If a causal curve is noth future and past inextendible then it is inextendible. There is a natural relation on events in spacetime based on whether they lie in each other's futures or pasts.

Definition 4.2.2 (Causal relations)**.**

1. *$p \ll q \iff \exists$ a future directed timelike curve from p to q*

2. *$p < q \iff \exists$ a non-trivial future directed causal curve from p to q*

We then say that q is in the timelike/causal future of p. The notation $p \leq q$ if q is used if q is either in the causal future of p or $p = q$. We can define the regions of causality as follows.

Definition 4.2.3 (Causal regions)**.**

1. *The **timelike future** of a point $p \in M$ is $I^+(p) := \{q \in M : p \ll q\}$*

2. *The **causal future** of a point $p \in M$ is $J^+(p) := \{q \in M : p \leq q\}$*

We defined the case for the future regions, it can be easily extended to past regions by reversing time. For example $J^-(p)$ would be set of all q such that $q \leq p$. For the rest of the discussion we may construct definitions for only future $(+)$ events, but let it be implicit that there are also time dual past $(-)$ definitions. Note that $I^+(p)$ is open $\forall p \in M$, but $J^+(p)$ may or may not be closed. —[EXAMPLE + PICTURE]—.

Proposition 4.2.1. *Let $p \in M$ be a point in spacetime and suppose $q \in J^+(p) - I^+(p)$, then any future directed causal curve from p to q must be a null geodesic.*

This is just the statement that no particle can move faster than the speed of light locally. From this proposition we have two more statements.

Remark 4.2.1. *The causal relations are transitive, and following is true:*

$$\forall p, q, r \in M \quad (p \ll q \ \wedge \ q < r) \ \vee \ (p < q \ \wedge \ q \ll r) \implies p \ll r.$$

We can generalise the causal/timelike futures of points to arbitrary subspaces in spacetime.

Definition 4.2.4. *Consider a subset $S \subseteq M$.*

1. *The **timelike future** of S is*

$$I^+(S) := \bigcup_{p \in S} I^+(p) = \{q \in M : p \ll q \quad \text{for some} \quad s \in S\}.$$

2. *The **causal future** of S is*

$$J^+(S) := \bigcup_{p \in M} J^+(p) = \{q \in M : p \leq q \quad \text{for some} \quad s \in S\}.$$

It will also be useful to consider causual relations only on open subsets in the following sense.

Definition 4.2.5. *Let $p \in M$. The **timelike future within** $U \subseteq M$, denoted $I^+(p, U)$, is the set of $q \in U$ such that there is a timelike curve within U such that $p \ll q$.*

Note, $I^+(p, U) \neq I^+(p) \cap U$ in general. Now in classical theories of physics we would like to have a sense of determinism.

4.3 Causality & Energy Conditions

In general relativity one often requires *causality conditions* on the underlying spacetime in order to prove strong results. One might reasonably rule out the ability to travel to ones own past for example. Such a trip would of course lead to various issues such as the famous 'grandfather paradox'. In spacetime these paths manifest as closed curves. If a massive observer can traverse such a curve then it must be timelike. So this brings us to our first proposed restriction to forbid observers from travelling back to their past.

Definition 4.3.1. *The **chronology condition** states there are no closed **timelike** curves in a spacetime M.*

This is already violated by compact spacetimes.

Proposition 4.3.1. *Every compact spacetime M contains closed timelike curves.*

Proof. The collection $\{I^+(p) : p \in M\}$ forms an open cover for M. Since M is compact there exists a finite subcover $\{I^+(p_1), \ldots, I^+(p_k)\}$ $k \in \mathbb{N}$. Without loss of generality, assume this cover has the minimal number of open sets. Since $\{I^+(p_i)\}_{i\in\mathbb{N}}^k$ is a cover, $p_1 \in I^+(p_j)$ for some j, so $I^+(p_1) \subseteq I^+(p_j)$. Now suppose $j \neq 1$, then we can remove $I^+(p_1)$ from the open cover, contradicting minimality. But $j = 1 \implies p_1 \in I^+(p_1) \implies \exists$ a closed timelike curve through p. $\square$

Timelike curves are not the only way to violate causality. A massive observer might be able to paradoxically send a laser back towards their past which kills themselves. Forbidding such pathologies leads to a slightly stronger condition.

Definition 4.3.2. *The **causality condition** states that there are no non-trivial closed **causal** curves in a spacetime M.*

This condition is still slightly to weak. It does not rule out the existence of 'almost closed' causal curves. —[PICTURES + NON-EXAMPLES]—. The most standard definition of causality forbids the existence of these pathologies.

Definition 4.3.3. *A spacetime M is **strongly causal at** $p \in M$ if there are arbitrarily small neighbourhoods $U \ni p$ such that any causal curve γ which leaves U with endpoints in U never returns to U. A spacetime is (everywhere) strongly causal if it strongly causal $\forall p \in M$.*

This is what will be needed for most of the discussion of singularity theorems.

Definition 4.3.4. *The **null energy condition** states that the energy-momentum tensor satisfies $T(X, X) \geq 0$ for all null vectors.*

Definition 4.3.5. *The **dominant energy condition** states that energy-momentum tensor satisfies $T(X, Y) \geq 0$ for all future directed vectors X, Y.*

There exist some classical fields which violate the null energy condition. —[REF]—. The dominant energy condition requires the 'energy flow' velocity to be less than the speed of light. This is more restrictive than the null energy condition, and is satisfied by most classical fields.

4.4 Important Subspaces

Traditionally in physics one considers having initial conditions on a 'subset of spacetime' with a differential equation. We expect to be able to determine what happens, and has happened, to the future and past of this subset by solving said differential equation with the prescribed initial values. To implement this in GR, where time becomes more complicated, we must restrict our definition of subset.

4.4.1 Achronal subsets

Definition 4.4.1. *A submanifold of spacetime, $\iota : \Sigma \hookrightarrow M$, is a* ***spacelike hypersurface*** *if $\iota^\star g$ has Riemannian signature.*

Consider 2-dimensional Minkowski spacetime with standard coordinates and metric

$$g = -\mathrm{d}t \otimes \mathrm{d}t + \mathrm{d}x \otimes \mathrm{d}x.$$

Define a family of submanifolds by,

$$\Sigma_\lambda \coloneqq \{(\lambda, x) : \lambda \in \mathbb{R}\},$$

which are just horizontal lines of constant time. Clearly each Σ_λ is a spacelike since the induced metric is just $\iota^\star g = \mathrm{d}x \otimes \mathrm{d}x$. This spacelike condition is still not enough implement determinism, we need something more.

Definition 4.4.2. *A subset Σ of spacetime is* ***achronal*** *if there exists no timelike curve connecting two arbitrary points $p, q \in \Sigma$.*

To see why this definition is needed, we come back to the trusty cylinder $\mathbb{R} \times S^1$ for a non-example. A family of one-dimensional submanifolds, Σ_λ, can be defined by,

$$t = \lambda\phi,$$

where (t, ϕ) are the standard coordinates on the cylinder and $\lambda \in \mathbb{R}$ is a constant. Visually these are lines wrapping around the cylinder infinitley many times. Now equip the space with the standard flat metric

$$g = -\mathrm{d}t \otimes \mathrm{d}t + \mathrm{d}\phi \otimes \mathrm{d}\phi.$$

For small λ, Σ_λ is a spacelike surface, approaching the circle for $\lambda \to 0$. However, Σ_λ is never achronal for $\lambda > 0$. This is clear when we consider the curve paramaterised by,

$$\gamma(\tau) = (\tau, 0),$$

for $\tau \in \mathbb{R}$. It's clear that the curve connects the points $(0, 0)$ and $(2\pi\lambda, 0)$ with a straight vertical line, which is evidently a timelike geodesic. In fact, the points $\{(n2\pi\lambda, 0) : n \in \mathbb{Z}\}$ are all connected by this geodesic. This poses an issue for specifying intitial conditions over Σ_λ since data at $(0, 0)$ can influence data $(2\pi\lambda, 0)$ in the future for example. Thus, it doesn't make sense to define initial conditions over spacelike hypersufaces if they are not also achronal.

Definition 4.4.3. *An* ***achronal boundary*** *is a subset of the form $\partial I^\pm(S)$ for some achronal subspace S.*

Proposition 4.4.1. *If an achronal boundary is non-empty, $\partial I^{\pm}(S) \neq \varnothing$, then it is a closed achronal C^0 hypersurface in M.*

Note that by C^0 hypersurface we mean a topological submanifold of codimension one. Achronal boundaries need not be smooth in general.

Definition 4.4.4. *Suppose $S \subseteq M$ is achronal. $p \in \bar{S}$ is an **edge point** of S if every neighbourhood $U \ni p$ contains a timelike curve, γ, from $I^-(p,U)$ to $I^+(p,U)$ that does not meet S, $\gamma \cap S = \varnothing$. Denote the set of edge points* $\mathrm{edge}(S)$. *We say that S is **edgeless** if* $\mathrm{edge}(S) = \varnothing$.

—-[GIVE TWO EXAMPLES + PICTURES]—.

Proposition 4.4.2. *An edgeless achronal set, $S : \mathrm{edge}(S) = \varnothing$, is a C^0 hypersurface.*

The proof of the above can be found in [O'N83].

4.4.2 Hyperbolic Spacetimes

A further refinement of the previous concept of achronality leads to a crucial definition.

Definition 4.4.5. *A **Cauchy surface** is an achronal spacelike hypersurface, $S \hookrightarrow M$, such that $p \in M - S \implies$ every inextendible causal path γ through p must intersect S : $\gamma \cap S \neq \varnothing$.*

The exsistence of such a surface is a strong condition on the underlying spacetime. We say that a spacetime is **globally hyperbolic** if it admits a Cauchy hypersurface. In globally hyperbolic spacetimes one can fully determine the future and past of initial conditions defined on the Cauchy surface. —[example of point removed from Minkowski]—.

Proposition 4.4.3. *A globally hyperbolic spacetime M contains no closed causal curves.*

Proof. Suppose for contradiction that there does exist a causal curve, γ , which repeats its trajectory. Now γ is clearly inextendible since its parameter can be thought of as repeating the curve's path infinitely. However, since M is globally hyperbolic, any closed causal curve which is inextendible must interest a Cauchy hypersurface S. Since the curve repeats, it must intersect this surface infinitely many times. Moreover, this surface is by definition achronal, leading to a contradiction since the causal curve provides a path between intersection points. □

A perhaps more standard definition of a hyperbolic spacetime can be stated directly from the causality conditions.

Definition 4.4.6. *A spacetime M is **globally hyperbolic** if:*

1. *M is strongly causal;*
2. *The sets $J^+(p) \cap J^-(q)$ are compact $\forall p, q \in M$.*

A pivotal property of hyperbolic spacetimes is that they are topologically simple.

Theorem 4.4.1. *Let M be a globally hyperbolic spacetime with Cauchy surface S. If S' is another Cauchy surface in M then it is homeomorphic to S, $S \cong_{\mathrm{Top}} S'$. Moreover, the spacetime is homoemorphic to the product $M \cong_{\mathrm{Top}} \mathbb{R} \times S$.*

—[give proof]—

Definition 4.4.7. *—[time function]—*

4.4.3 Domains of Dependence

Definition 4.4.8. *Let S be an achronal set in a spacetime M. The* ***future domain of dependence*** *of S is the set $D^+(S)$ of all points $p \in M$ such that every past inextendible causal curve from p meets S.*

Likewise, one can define the **past domain of dependence** $D^-(S)$ for an achronal set by taking the time dual of the previous definition. Physically, infomation travels along causal curves. So a point in $D^+(S)$ only recieves information from S; the future domain of dependence is the region where events are predictable from data on S. We write $D(S) \coloneqq D^+(S) \cup D^-(S)$ for the **(total) domain of dependence** for an achronal set S.

Definition 4.4.9. *Let S be an achronal set in a spacetime M. The* ***future Cauchy horizon*** *of S is the future boundary of the future domain of dependence $D^+(S)$. Explicitly:*

$$H^+(S) \coloneqq \{p \in \overline{D^+(S)} : I^+(p) \cap D^+(p) = \varnothing\}$$

Again one can take the time dual of this definition to define the **past Cauchy horizon** $H^-(S)$. The **(total) Cauchy horizon** is denoted $H(S) \coloneqq H^+(S) \cup H^-(S)$. Then it follows that $\partial D(S) = H(S)$. Cauchy surfaces have a nice characterisation in terms of domains of dependence and Cauchy horizons.

Proposition 4.4.4. *Let M be a spacetime and S an acausal set. The following satements are equivalent.*

1. *S is a Cauchy surface for M;*
2. *$D(S) = M$;*
3. *$H(S) = \varnothing$.*

4.5 Inner Product Structure on the Tangent Space

As an aside, let's recall some facts about inner products with indefinite signature. For this section we fix a given metric on the tangent space, (T_pM, g).

Definition 4.5.1. *The orthognal complement of a vector subspace, $V \subseteq T_pM$, is defined by*

$$V^\perp \coloneqq \{X \in T_pM : g(X, Y) = 0, \forall Y \in V\}.$$

Proposition 4.5.1. *Let $V \subseteq T_pM$ be a subspace. The following statements are equivalent:*

$$g\big|_V \text{ is non-degenerate} \iff V \cap V^\perp = \{0\} \iff T_pM = V \oplus V^\perp.$$

There are two inequalities which always hold for Lorentzian metrics.

Proposition 4.5.2.

1. *The* ***reversed Schwarz inequality****,*

$$|g(X, Y)| \geq |X||Y|, \tag{4.2}$$

holds for all causal vectors X and Y.

2. *The* ***reversed triangle inequality****,*

$$|X+Y| \geq |X| + |Y|, \tag{4.3}$$

holds for all similary directed causal vectors X, Y.

Proof. First we will show the reversed Schwarz inequality, which then implies the reverse triangle inequality. The inequality is trivially satisfied for null vectors since all terms are zero. Assuming X is timelike, we consider the subspace

$$\mathbb{R} \cdot X \coloneqq \operatorname{span}_{\mathbb{R}} X = \{\lambda X : \lambda \in \mathbb{R}\},$$

and then apply the decomposition from proposition 4.5.1,

$$T_p M = \mathbb{R} \cdot X \oplus (\mathbb{R} \cdot X)^{\perp},$$

to write $Y = \lambda X + Y^{\perp}$. Here $Y^{\perp} \in (\mathbb{R} \cdot X)^{\perp}$ is spacelike since X is timelike. Now we have,

$$g(X,Y) = g(X, \lambda X + Y^{\perp}) = \lambda g(X,X) + g(X, Y^{\perp}) = \lambda g(X,X), \tag{4.4}$$

where $g(X, Y^{\perp}) = 0$ by definition. Also note that,

$$g(Y,Y) = \lambda^2 g(X,X) + 2g(X,Y^{\perp}) + g(Y^{\perp}, Y^{\perp}) = \lambda^2 g(X,X) + g(Y^{\perp}, Y^{\perp}), \tag{4.5}$$

again using the definition of $Y^{\perp}$. Now square both sides of eq. (4.4) to get,

$$g(X,Y)^2 = g(X,X)\lambda^2 g(X,X) = g(X,X)\left(g(Y,Y) - g(Y^{\perp}, Y^{\perp})\right), \tag{4.6}$$

where we substituted eq. (4.5) in the second step. Finally remember that $g(X,X) < 0$, $g(Y,Y) \leq 0$, and $g(Y^{\perp}, Y^{\perp}) > 0$. Then the inequality follows,

$$\begin{aligned} & g(X,Y)^2 \geq g(X,X)g(Y,Y) \\ \implies & |g(X,Y)| \geq |X||Y|, \end{aligned}$$

by taking absolute values and realisng square roots can be applied to both positive sides of the inequality. For the reveresed triangle inequality we recall that if X and Y are similarly directed causal then $g(X,Y) \leq 0$. Then keeping track of signs,

$$\begin{aligned} |X+Y|^2 &= -g(X+Y, X+Y) = -g(X,X) - 2g(X,Y) + g(Y,Y) \\ |X+Y|^2 &= |X|^2 + 2|g(X,Y)| + |Y|^2, \end{aligned}$$

and then finally use the reveresed Schwarz inequality to get,

$$\begin{aligned} & |X+Y|^2 \geq |X|^2 + 2|X||Y| + |Y|^2 = (|X| + |Y|)^2 \\ \implies & |X+Y|^2 \geq |X|^2 + |Y|^2. \end{aligned}$$

□

These counter-intuitive relatations are the cause of the famous 'twin paradox' from special relativity.

4.6 Defining Singularities

—[discuss curvature invariants + excised points]—

The discussion in this section follows [HE23] mainly. In differential geometry one can construct a distance function $d_g(x, y)$ on a Riemannian manifold (M, g). This is computed using the usual definition of lengths and geodesics. On the other hand, in point set topology, one has a notion of a metric space. A priori this need not arise from a Riemannain metric. We say that a Riemannian manifold is geodesically complete if every Cauchy sequence converges with respect to the distance function. On the other hand, for a metric space (M, d), we say it is metric complete analagously if it obeys a similar condition for its own distance function. On a Riemannian manifold, with a positive-definite metric, one can show that the two notions are equivalent. However, in the Lorentzian case, it becomes more complicated. One has to distinguish between—[complete]—.

Definition 4.6.1 (Singularity)**.** *A spacetime M is **singular** if it is timelike or null geodesically incomplete.*

4.7 Null Geometry

Definition 4.7.1. *Let M be a spacetime. A **null hypersurace** is a smooth, connected hypersurface $\iota : S \hookrightarrow M$ such that $\iota^\star g$ is degenerate.*

In particular:

$$\forall p \in S, \quad \exists 0 \neq K_p \in T_pS : (\iota^\star g)(K_p, X_p) = 0 \quad \forall X_p \in T_pS$$

It follows immediately that K_p, called the **direction of degeneracy**, is null $\langle K_p, K_p \rangle = 0$. Furthermore, we have $(\mathbb{R} \cdot K_p)^\perp = T_pS$. Without loss of generality we can take K_p to be future pointing with respect to the time-orientation on M. Any non-zero vector that isn't a multiple of K_p is also necessarily spacelike.

$$0 \neq X_p \in T_pS : X_p \notin \mathbb{R} \cdot K_p \implies (\iota^\star g)(X_p, X_p) > 0$$

Thus these vectors can be stitched together to form global section, $K \in \mathfrak{X}(S)$, on the hypersurface. Note that this section is indeed a smooth vector field when the K_p are suitably normalised. K is also unique up to pointwise multiplication by a postive function. In Minkowski space there are the future and past achronal boundaries, $\partial I^\pm(p)$, of a point $p \in \mathbb{R}^{3,1}$. These form the future and past light cones at p, and neither are smooth. Away from the vertex however, $\partial I^\pm(p) - \{p\}$, are both null hypersurfaces. The notation $\langle -, - \rangle \coloneqq (\iota^\star g)(-, -)$ may be used when the context is clear.

Proposition 4.7.1. *Suppose $\iota : S \hookrightarrow M$ is a null hypersurface. Then the integral curves of a null vector field, $K \in \mathfrak{X}(S)$, are null geodesics under a suitable choice of parameterisation.*

Proof. We would like to show that $\nabla_K K = \lambda K$ for some $\lambda \in \mathbb{R}$. Then the integral curves of K are pregeodesics, and hence geodesics once reparemeterised appropriately. Consider an arbitrary vector $X_p \in T_pS$. Then this can be extended as an invariant vector field, $X \in \mathfrak{X}(\gamma)$, along an integral curve γ of K which passes through the point $p \in S$.

$$\mathcal{L}_K X = \nabla_K X - \nabla_X K = 0$$

By definition of the null vector field, $\langle K, X\rangle_\gamma = 0$, since X is still tangent to S. Now differentiate,

$$\begin{aligned} K\langle K, X\rangle &= \langle \nabla_K K, X\rangle + \langle K, \nabla_K X\rangle = 0 \\ \implies \langle \nabla_K K, X\rangle &= -\langle K, \nabla_X K\rangle, \end{aligned}$$

where metric compatiblilty was used to introduce the covariant derivative. Note that on the other hand,

$$X\langle K, K\rangle = \langle \nabla_X K, K\rangle + \langle K, \nabla_X K\rangle = 2\langle K, \nabla_X K\rangle,$$

by symmetry. Then, using the defintion of the invariant extenstion of X, we obtain

$$\langle \nabla_K K, X\rangle = -\frac{1}{2} X\langle K, K\rangle = 0.$$

So we have shown that $\mathbb{R} \cdot \nabla_K K \perp T_p S$. In particular, by the uniqueness of the null vector field up to pointwise scaling,

$$\langle \nabla_K K, X\rangle\big|_p = 0 \quad \forall X \implies \nabla_K K = \lambda K.$$

□

In Riemannian geometry it's often useful to study submanifolds through the normal vector field associated to it. Similarly, we can study the geometry of null hypersurfaces through the null vector field. It's conveniant to work in a formalism where the direction of degeneracy is quotiented out.

Definition 4.7.2. *Let $\iota : S \hookrightarrow M$ be a null hypersurface.*

1. *Define an equivalence relation on the tangent space, $T_p S$, by*

$$X, X' \in T_p S, \quad X \sim_K X' \iff X - X' = \lambda K, \quad \lambda \in \mathbb{R}.$$

2. *Denote the equivalence class by $\bar{X} \coloneqq [X]_{\sim_K} = \{Y \in T_p S : X \sim_K Y, X \in T_p S\}$. The **mod K tangent space** at a point $p \in S$ is the quotient vector space*

$$T_p S/K \coloneqq \{\bar{X} : X \in T_p S\}.$$

3. *The **mod K tangent bundle** is the disjoint union of the mod K tangent spaces over S.*

$$TS/K \coloneqq \bigsqcup_{p \in S} T_p S/K$$

Recall we write $d = \dim M$, and the hypersurface is codimension 1, $\dim S = d - 1 \coloneqq n$. Then the mod K tangent bundle is a smooth vector bundle of rank $n - 1$. Note that, despite the name, the mod K tangent bundle does not depend on the choice of null vector field.

Proposition 4.7.2. *The mod K tangent bundle is well-defined for an arbitray choice of null vector field. In particular, for two null vector fields $K, K' \in \mathfrak{X}(S)$:*

$$TS/K \cong TS/K'$$

Proof. It suffices to show that the quotient tangent spaces are equal, $T_pS/K = T_pS/K'$, $\forall p \in S$. Suppose $X, X' \in T_pS/K$ then,

$$X - X' = \alpha K,\ \alpha \in \mathbb{R},$$

by the previous definition. Recall on the other hand that null vector fields on a null hypersuface are unique up to pointwise scaling by a non-zero number. So if $K \in T_pS$ is a null vector at p, $\exists \beta \in \mathbb{R}_{>0} : K = \beta K'$. Then it is immediate that

$$X - X' = \alpha K \iff X - X' = \alpha\beta K'.$$

So $X \sim_K X' \iff X \sim_{K'} X'$, and $T_pS/K = T_pS/K'$. □

There is a natural metric induced by the quotient map.

Definition 4.7.3. *Let $\iota : S \hookrightarrow M$ be a null hypersurface with K the direction of degeneracy. The **(null) induced metric**, $h : T_pS/K \otimes T_pS/K \longrightarrow \mathbb{R}$, is defined by*

$$h(\bar{X}, \bar{Y}) \coloneqq \langle X, Y \rangle.$$

Proposition 4.7.3. *The induced metric is well-defined mod K.*

Proof. Consider two vectors, $X, Y \in T_pS$, so that $X' = X + \alpha K$ and $Y' = Y + \beta K$ are two different representatives in the quotient. Then,

$$\begin{aligned} h(X', Y') &= \langle X + \alpha K, Y + \beta K \rangle \\ &= \langle X, Y \rangle + \langle X, \beta K \rangle + \langle \alpha K, Y \rangle + \langle \alpha K, \beta K \rangle \\ &= \langle X, Y \rangle. \end{aligned}$$

□

Definition 4.7.4. *The **null Weingarten map**, $b_K : T_pS/K \longrightarrow T_pS/K$, is defined by:*

$$b_K(\overline{X}) \coloneqq \overline{\nabla_X K}$$

Proposition 4.7.4.

1. *The null Weingerten map is well-defined.*
2. *$b_K = b$ is unique up to a smooth positive function.*

Proof.

1. Consider two equivalent representatives, $X, X' \in T_pS : X' = X + \alpha K$.

$$\begin{aligned} b_K(\overline{X'}) &= \overline{\nabla_{X'} K} = \overline{\nabla_{X+\alpha K} K} \\ &= \overline{\nabla_X K} + \alpha \overline{\nabla_K K} \\ &= \overline{\nabla_X K} = b_K(X) \end{aligned}$$

   In the second line we have used the earlier result that the integral curves of K are null (pre)geodesics, $\nabla_K K = \lambda K$, $\lambda \in \mathbb{R}$.

2. If $\tilde{K} \in \mathfrak{X}(S)$ is another null vector field then $\exists f \in C^\infty(S)$, a smooth positive function, such that $\tilde{K} = fK$. Then $\forall X \in T_pS$,

$$\begin{aligned} b_{\tilde{K}}(\overline{X}) &= \overline{\nabla_X(fK)} \\ &= \overline{f\nabla_X K} + \overline{X(f)\cdot K} \\ &= f b_K(\overline{X}). \end{aligned}$$

□

From now on the null vector field associated to the null Weignarten map will be omitted in the notation.

Definition 4.7.5. *The **null second fundamental form**, $B : T_pS/K \otimes T_pS/K \longrightarrow \mathbb{R}$, associated to a null vector field K on S is defined by*

$$B(\overline{X}, \overline{Y}) \coloneqq h(b(\overline{X}), \overline{Y}).$$

Proposition 4.7.5. *The null second fundamental form is symmetric. Equivalently the null Weingarten map, b, is self-adjoint with respect to the induced metric h.*

Proof. Consider two vectors $X, Y \in T_pS$ and extend them to vector fields in a neighbourhood of p. Then,

$$\begin{aligned} \langle K, Y\rangle = 0 \quad &\& \quad \langle K, X\rangle = 0 \\ \implies X\langle K, Y\rangle = 0 \quad &\& \quad Y\langle K, X\rangle = 0 \\ \iff \langle \nabla_X K, Y\rangle = -\langle K, \nabla_X Y\rangle \quad &\& \quad \langle \nabla_Y K, X\rangle = -\langle K, \nabla_Y X\rangle. \end{aligned}$$

Recall that the Levi-Civita connection is torsion free,

$$\nabla_X Y - \nabla_Y X = [X, Y].$$

Now apply the definition of the fundamental form.

$$\begin{aligned} B(\overline{X}, \overline{Y}) = h(b(\overline{X}), \overline{Y}) &= \langle \nabla_X K, Y\rangle = -\langle K, \nabla_X Y\rangle \\ &= -\langle K, \nabla_Y X\rangle - \langle K, [X, Y]\rangle \\ &= \langle \nabla_Y K, X\rangle = \langle X, \nabla_Y K\rangle \\ &= h(\overline{X}, b(\overline{Y})) = B(\overline{Y}, \overline{X}) \end{aligned}$$

□

Definition 4.7.6. *A null hypersuface is **totally geodesic** if $B \equiv 0$.*

If S is totally geodesic, then any geodesic starting tangent to S stays in S. Null hyperplanes in Minkowski spacetime are totally geodesic. Another example is in Schwarzschild spacetime. The event horizon is a null hypersurface, and light rays tangent to it must stay tangent. Studying the deviation of light rays is a useful tool to analyse the geometry of spacetimes. One can define a function which formalises this notion.

Definition 4.7.7. *Let b be the null Weingarten map for a null hypersurface. The* ***null mean curvature*** *or* ***null expansion scalar****, $\vartheta \in C^\infty(S)$, is defined by*

$$\vartheta := \operatorname{Tr} b. \tag{4.7}$$

Consider a spacelike hypersurface in M which is transverse to K at $p \in S$. Denote the intersection of the hypersuface with S by Σ. This is a codimension 2 spacelike submanifold such that K is orthognal on it. —[DRAW PICTURE]—. Now choose an orthonormal basis $\{\sigma_\alpha\}_{\alpha=1}^{n-1}$ of $T_p\Sigma$. Passing to the quotient we have an orthonormal basis, $\{\overline{\sigma}_{\alpha=1}^{n-1}\}$, for T_pS/K. Then the null mean curvature is,

$$\vartheta = \sum_{\alpha=1}^{n-1} h(b(\overline{\sigma}_\alpha), \overline{\sigma}_\alpha) = \sum_{\alpha=1}^{n-1} \langle \nabla_{\sigma_\alpha} K, \sigma_\alpha \rangle, \tag{4.8}$$

which is recognisable as the **divergence** of the null vector field along Σ. If ω is the volume form on Σ then we can write,

$$\mathcal{L}_K \omega = \vartheta \omega, \tag{4.9}$$

in the direction of the flow generated by K. At a point $p \in \mathbb{R}^{3,1}$ in Minkowski spacetime, consider the future null cone in Minkowski spacetime away from the vertex $\partial I^+ - \{p\}$. The null expansion on this hypersurface is greater than zero, $\vartheta > 0$, matching the fact that the null generators are diverging in time. Likewise, for the past null cone, $\partial I^- - \{p\}$, we have $\vartheta < 0$ since the null generators are converging to p.

Recall from Proposition 4.7.4 that $b_{\tilde{K}} = f b_K$ for some other null vector field, $\tilde{K}$, related by a positive smooth function $f \in C^\infty(S)$ such that $\tilde{K} = fK$. It follows immediately, from the linearity of the trace operation, that $\tilde{\vartheta} = f\vartheta$. In particular the sign of the null expansion is preserved. This is useful since inequalities such as $\vartheta > 0$ do not depend on the choice of null vector field. Now let's define these constructions over curves. Consider an affinely parametrised null geodesic generator, $s \mapsto \gamma(s)$, on S. A vector field, $s \mapsto \mathcal{Y}(s) \in T_{\gamma(s)}S/K$, is smooth if $\exists$ a smooth vector field, $s \mapsto Y \in T_{\gamma(s)}S$, such that $\mathcal{Y}(s) = \overline{Y(s)}$. Now we can differentiate the vector field on the mod K bundle by defining

$$\mathcal{Y}'(s) := \overline{\nabla_{\dot{\gamma}} Y(s)}, \tag{4.10}$$

where $\nabla_{\dot{\gamma}} = \gamma^\star \nabla$ is the induced covariant derivative.

The null Weignarten map can now be defined along the curve in the usual way, $b(s) := b_{\dot{\gamma}(s)} : T_{\gamma(s)}S/\dot{\gamma}(s) \longrightarrow T_{\gamma(s)}S/\dot{\gamma}(s)$. The derivative of the null Weingarten map is defined by,

$$b'(\overline{X}) := \nabla_{\dot{\gamma}} b(\overline{X}) - b(\overline{\nabla_{\dot{\gamma}} X}). \tag{4.11}$$

Proposition 4.7.6. *The null Weingarten map satisfies the Ricatti equation,*

$$b' + b \circ b + R = 0, \tag{4.12}$$

where $R : T_{\gamma(s)}S/\dot{\gamma}(s) \longrightarrow T_{\gamma(s)}S/\dot{\gamma}(s)$ is the Riemann curvature endomorphism defined by,

$$R(\overline{X}) := \overline{R(X, \dot{\gamma}(s))\dot{\gamma}(s)}. \tag{4.13}$$

Proof. For $s \in (a, b)$, let $p = \gamma(s_0)$. The null vector field, K, on S can be scaled in a neighbourhood $U \ni p$ so that K is an affinely parameterised geodesic and $K|_\gamma$ is the velocity vector field to γ. Explicitly, $K_{\gamma(s)} = \dot{\gamma}(s)$. Now for $X \in T_pS$ we can pick another neighbourhood $V \ni p$ in which X is extended to smooth vector field such that

$$[X, K] = \nabla_X K - \nabla_K X = 0,$$

on the intersection $U \cap V$. Now the Riemann curvature simplifies,

$$\begin{aligned} R(X, K)K &= \nabla_X \nabla_K K - \nabla_K \nabla_X K - \nabla_{[X,K]} K \\ &= -\nabla_K \nabla_X K = -\nabla_K \nabla_K X. \end{aligned}$$

In the first line $\nabla_K K = 0$ by the null geodesic property. The Lie bracket $[X, K]$ vanishes to eliminate the third term in the first line, and leads to the final line. So along the curve,

$$\nabla_{\dot{\gamma}} \nabla_{\dot{\gamma}} X = -R(X, \dot{\gamma})\dot{\gamma},$$

meaning that X is a **Jacobi field** along γ. Turning to the derivative of the Weingarten map,

$$\begin{aligned} b'(\overline{X}) &= \nabla_{\dot{\gamma}} \overline{\nabla_X K} - b(\overline{\nabla_K X}) \\ &= \nabla_{\dot{\gamma}} \overline{\nabla_K X} - b \circ b(\overline{X}) \\ &= \overline{\nabla_{\dot{\gamma}} \nabla_{\dot{\gamma}} X} - b \circ b(\overline{X}) \\ &= -\overline{R(X, \dot{\gamma})\dot{\gamma}} - b \circ b(\overline{X}). \end{aligned}$$

□

Taking the trace of the Riccati equation leads to,

$$\dot{\vartheta} = -\mathrm{Rc}(\dot{\gamma}, \dot{\gamma}) - \sigma^2 - \frac{1}{n-1}\vartheta^2, \tag{4.14}$$

where $\sigma := \sqrt{\mathrm{Tr}\, \hat{b}^2}$ with $\hat{b} = b - \frac{\vartheta}{n-1}\,\mathrm{id}$. The above is a form of the **Raychaudhuri equation** for null irrotational geodesic congruence. σ is sometimes called the **shear scalar** and $\hat{b}$ is the traceless part of the Weingarten map.

Proposition 4.7.7. *Suppose a spacetime, M, obeys the null energy condition $Rc(X, X) \geq 0$. Let $S \hookrightarrow M$ be a smooth null hypersurface such that the generators of S are future geodesically complete. Then S has non-negative null expansion $\vartheta \geq 0$.*

Proof. Suppose for contradiction $\exists p \in S : \vartheta < 0$. Pick an affinely paramaterised null geodesic generator, $s \mapsto \gamma(s)$, passing through $p = \gamma(0)$. Then $\vartheta(0) < 0$. The null energy condition applied to the Raychaudhuri equation gives an inequality,

$$\frac{\mathrm{d}\vartheta}{\mathrm{d}s} \leq -\frac{1}{n-1}\vartheta^2. \tag{4.15}$$

But $\vartheta^2 > 0 \implies \dot{\vartheta} < 0, \quad \forall s > 0$. Rearranging,

$$-\frac{1}{\vartheta^2}\frac{\mathrm{d}\vartheta}{\mathrm{d}s} \geq \frac{1}{n-1}, \tag{4.16}$$

and noticing that,

$$\frac{\mathrm{d}}{\mathrm{d}s}\left(\frac{1}{\vartheta}\right) = -\frac{1}{\vartheta^2}\frac{\mathrm{d}\vartheta}{\mathrm{d}s},$$

results in

$$\frac{\mathrm{d}}{\mathrm{d}s}\left(\frac{1}{\vartheta}\right) \geq \frac{1}{n-1}. \tag{4.17}$$

So $1/\vartheta \to 0 \iff \vartheta \to -\infty$ in finite affine time. This contradicts smoothness. □

4.8 Penrose's Theorem

—[Witten lectures Schwarzschild motivating example]—

Consider a closed (compact without boundary), spacelike submanifold $\iota : \Sigma \hookrightarrow M$. $\forall p \in \Sigma$, $(T_p\Sigma)^\perp$ is timelike and 2-dimensional. There are then two future directed null directions orthogonal to Σ. If the normal bundle to Σ is trivial, then there two non-vanishing null normal vector fields which are unique up to scaling. Denote the null vectors associated to these two directions by ℓ_- and ℓ_+ for ingoing and outgoing light rays. Denote the associated second fundamental forms by

$$\chi_\pm(X,Y) \coloneqq g(\nabla_X \ell_\pm, Y), \tag{4.18}$$

and the expansions scalars are the usual traces

$$\vartheta_\pm \coloneqq \mathrm{Tr}_{\iota^\star g}\, \chi_\pm = (\iota^\star g)^{ab}(\chi_\pm)_{ab} = \mathrm{div}_\Sigma\, \ell_\pm. \tag{4.19}$$

Definition 4.8.1. *A **trapped surface** is a closed spacelike submanifold such that both $\vartheta_+ < 0$ and $\vartheta_- < 0$.*

Now the famous result by Penrose can be formulated [Pen65].

Theorem 4.8.1 (Penrose)**.** *Let M be a globally hyperbolic spacetime satisfying the null energy condition, $Rc(X,X) \geq 0$ for all null vectors X. Suppose M has a non-compact Cauchy surface S. If M contains a trapped surface, Σ, then M is future null geodesically incomplete.*

Proof.

- *Claim: $\partial I^+(\Sigma)$ is non-compact.*
  Suppose for contradiction that $\partial I^+(\Sigma)$ is in fact compact. Since $\partial I^+(\Sigma)$ is an achronal boundary, it is an achronal C^0 hypersurface by Proposition 4.4.1. However, since M is globally hyperbolic, every Cauchy surface is homeomorphic $\partial I^+(\Sigma) \cong_{\mathrm{Top}} S$. This contradicts the assumption of a non-compact Cauchy hypersurface.

- *Claim: There exists a future inextendible null geodesic in $\partial I^+(\Sigma)$.*
  From point set topology we have

$$\partial I^+(\Sigma) \cong \overline{I^+(\Sigma)} - \mathrm{int}\, I^+(\Sigma).$$

Moreover, since the spacetime is globally hyperbolic,

$$\partial I^+(\Sigma) \cong J^+(\Sigma) - I^+(\Sigma).$$

From Proposition 4.2.1 each $q \in \partial I^+(\Sigma)$ must lie on a null geodesic in $\partial I^+(\Sigma)$ with the initial endpoint on $p \in \Sigma$. Furthermore, one can show that this null geodesic must meet Σ orthogonally [O'N83]. Since $\partial I^+(\Sigma)$ is closed and non-compact, $\exists \{q_k\}_{k\in\mathbb{N}} \subset \partial I^+(\Sigma)$, a sequence of points which diverge to infinity. Now, $\forall k \in \mathbb{N}$, there exists a null geodesic, γ_k, from Σ to q_k which is contained in $\partial I^+(\Sigma)$ and meets Σ orthogonally. Compactness of Σ implies that every subsequence $\{p_k\}_{k\in\mathbb{N}} \subset \Sigma$ has a convergent subsequence $p_{k_j} \to p \in \Sigma$. So for the sequence of geodesics $\{\gamma_k\}_{k\in\mathbb{N}}$ there exists a subsequence $\gamma_{k_j} \to \gamma$ convergent to an inextendible null geodesic in $\partial I^+(\Sigma)$ meeting Σ orthogonally at p.

- *Claim: γ is future incomplete.*
  Suppose for contradiction that γ is in fact future complete. The achronality of $\partial I^+(\Sigma)$ implies that no other future directed null geodesic starting on Σ can meet γ. Then there are no null focal points to Σ along γ, and so there exists a null hypersurface $H \subset \partial I^+(\Sigma)$ containing γ [O'N83]. Let ϑ be the null expansion of H along γ. By assumption there exists a trapped surface, Σ, and so $\vartheta(p) < 0$. From similar arguments as in Proposition 4.7.7, $\vartheta \to -\infty$ in finite affine time. This is a contradiction, hence γ must be incomplete.

□

—[include more details in proof]— The assumption of a non-compact Cauchy surface is motivated by the usual spacetimes considered in physics which satisfy asymptotic boundary conditions. For example, the Schwarzschild black hole has the property of being asymptotically flat. It is a common 'folk statement' that Penrose's theorem guarantees that black holes are formed under gravitational collapse. This is not strictly true. What Penrose's theorem says is about trapped surfaces which are *already* behind a horizon. Moreover, there are subtle differences between singularities in the traditional sense of curvature divergence, and singularities as characterised by curve incompleteness. Penrose's theorem, and other singularity theorems, apply to the latter kind. This is why some authors prefer to refer to them as 'incompleteness theorems'.

4.9 Cosmic Censorship

—[This section is under construction]— When singularities form in spacetime they often seem to be accompanied by an event horizon. For example the Schwarzchild horizon is 'clothed' by a horizon, so as to never be observed directly in nature. On the other hand, for Kerr spacetime, the event horizon can be stripped away to reveal a 'naked singularity' which is no longer causally disconnected from observers. This happens for large values of the spin parameter beyond the extremal case. However, these kinds of singularites have not been observed, and it is perhaps reasonable to suspect that there are laws in place which prevent such occcurences. This is idea is attempted to be made precise through the 'weak cosmic censorship hypothesis'. On the other hand there is also the issue of 'losing predictability' in singular spacetimes. The 'strong cosmic censorship hypothesis' states roughly that a generic solution of Einstein's field equations cannot be continued beyond the Cauchy horizon.

The censorship problems are two inceridbly importent open problems in General Relativity. Even defining them precisely is tricky. There are roughly two different approaches, one such was

the original idea by Penrose, while the more modern formulation is to do with the theory of partial differential equations. Note also that, despite the naming, the stong and weak cosmic censorship conjectures are logically independent. Let us recall an important theorem and definitions therein which will be necessary for the upcoming discussion.

Theorem 4.9.1 (Maximal future Cauchy developement)**.** *Let Σ be a 3-dimensional Riemannian manifold with metric $h \in \mathrm{Sym}^2 T^\star\Sigma$. Consider another symmetric tensor field $K \in \mathrm{Sym}^2 T^\star\Sigma$. Suppose that the* ***initial data*** *(Σ, h, K) satisfies the* ***constraint equations****:*

$$D_b K^b_a - D_a K^b_b = 0 \tag{4.20}$$

$$R_h + (K^a_a)^2 - K_{ab}K^{ab} = 0 \tag{4.21}$$

Here D and R_h are the covariant derivative and Ricci scalar associated to h. Then there exists a unique, up to isometry, 4-dimensional spactime (M, g) such that:

1. *(M, g) is a solution to Einstein's field equations.*
2. *(M, g) is globally hyperbolic with Cauchy surface $\iota : \Sigma \hookrightarrow M$.*
3. *$h = \iota^\star g$ and K are the induced metric and extrinsic curvature on Σ.*
4. *Any other spacetime (M', g') satisfying (1.–3.) can be mapped isometrically into a subset of (M, g).*
5. *The solution g depends continuously on the inital data (h, K) on Σ.*

—[Discuss conformal/Penrose diagrams in another section, define these formally]—. In asymptotically flat spacetimes we have a few definitions which will be useful. For Schwarzschild spacetime in the usual coordinates we say that **future null infinity** $\mathscr{I}^+$ is characterised by $r \to \infty$ and $t \to \infty$. Similarly, **past null infinity** is denoted $\mathscr{I}^-$ for $t \to -\infty$. Using these definitions one can construct a mathematical definition of a black hole.

Definition 4.9.1. *Let M be a causal and asymptotically flat spacetime. M contains a* ***black hole*** *if M is not contained in $J^-(\mathscr{I}^+)$. The* ***black hole region*** *is defined as*

$$B \coloneqq M - J^-(\mathscr{I}^+), \tag{4.22}$$

and the ***event horizon*** *is the boundary in spacetime*

$$H \coloneqq J^-(\mathscr{I}^+) \cap M. \tag{4.23}$$

—[define naked singularity]—

Proposition 4.9.1 (Weak cosmic censorship conjecture)**.** *Let M be a strongly causal spacetime which asymptotically flat at null infinity. The following statements are euivalent.*

1. *M contains no 'naked singularities' with respect to $J^-(\mathscr{I}^+)$.*
2. *The spacetime $J^-(\mathscr{I}^+)$ is globally hyperbolic.*

Proposition 4.9.2 (Strong cosmic cencorship conjecture)**.** *Consider intial data (Σ, h, K) with either Σ compact or asymptoically flat. The maximal globally hyperbolic developement is inextendible.*

4.10 Instability of Kaluza-Klein Theories

Following from Penrose's theorem one can deduce that 'tame' spacetimes prefer to be topologically trivial when possible. The result is due to [Gan75] and [Lee76].

Theorem 4.10.1. *Let M be a globally hyperbolic spacetime satisfying the null energy condition and admitting an asymptotically flat spacelike Cauchy surface Σ. If $\pi_1(\Sigma) \neq 0$ then M is future null geodesically incomplete.*

—[discuss moduli in string theory for example]—

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